



Predicting Credit Risk Outcomes in Loan Portfolios: A Data-Driven Alternative to Default-Based Credit Risk Models

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Certification Page

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ABSTRACT

Credit risk assessment is central to lending decisions, portfolio management, and financial stability. Traditional credit risk models typically focus on default prediction and may not fully capture the varying levels of borrower risk that exist before default occurs. This study explores the use of machine learning as an alternative approach to predicting borrower credit risk outcomes.

A quantitative research design was adopted using the Lending Club loan dataset, containing approximately two million anonymised consumer loan records issued between 2007 and 2018. The study employed a reduced-form modelling framework in which an empirical transition matrix was developed as a benchmark for assessing credit risk changes, while a Random Forest model was used to predict borrower credit risk outcomes. Model performance was improved through cross-validation and hyperparameter optimisation and evaluated using standard classification metrics. SHAP analysis was applied to enhance model interpretability and transparency.

The results show that the optimised Random Forest model achieved strong predictive performance, with an overall accuracy of 79% and particularly high effectiveness in identifying adverse credit outcomes. The findings also revealed the most important factors influencing borrower risk. In addition, the transition matrix analysis confirmed a clear relationship between initial credit quality and final loan performance.

The study concludes that machine learning can provide a more data-driven, informative and flexible approach to credit risk assessment than traditional default-focused methods. By combining predictive modelling with explainable artificial intelligence, the proposed framework offers a practical tool for improving borrower risk identification and supporting more proactive credit risk management. It is recommended that financial institutions explore the integration of explainable machine learning models into credit risk monitoring and portfolio management.

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DEDICATION

This dissertation is dedicated to my family for their unwavering love, support, and encouragement throughout my academic journey.

It is also dedicated to my data science tutors, whose guidance strengthened my analytical thinking, and to my colleagues and mentors within the Enterprise Risk Management team at Sterling Holding Company, whose practical insights inspired this study.

Finally, this work is dedicated to all professionals and researchers advancing the application of data science in financial risk management.

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List of Abbreviations

Abbreviation		Full Term
AI	-	Artificial Intelligence
BCBS	-	Basel Committee on Banking Supervision
BIS	-	Bank for International Settlements
CBN	-	Central Bank of Nigeria
EAD	-	Exposure at Default
F1	-	F1 Score
GDP	-	Gross Domestic Product
IFRS	-	International Financial Reporting Standards
K-Fold CV	-	K-Fold Cross-Validation
LGD	-	Loss Given Default
MLR	-	Multinomial Logistic Regression
MC	-	Markov Chain
ML	-	Machine Learning
NPL	-	Non-Performing Loan
PD	-	Probability of Default
SHAP	-	Shapley Additive Explanations
XAI	-	Explainable Artificial Intelligence
GridSearchCV	-	Grid Search Cross-Validation
LIME	-	Local Interpretable Model-Agnostic Explanations
PDP	-	Partial Dependence Plots
ICE	-	Individual Conditional Expectation

CHAPTER 1: Introduction

1.1 Background of the Problem

Credit risk is defined as the possibility that a borrower fails to meet debt repayment obligations. It is one of the most significant sources of financial risk in financial institutions. Banks and lending institutions allow large portions of their funds to be used for credit-related assets like corporate loans, consumer lending, and debt securities, exposing them to both risks related to individual borrowers and risks caused by wider economic conditions.

According to the Bank for International Settlements (BIS, 2023), total global credit to the non-financial sector was more than USD 160 trillion, highlighting how large and important credit exposure has become in the global economy. Similarly, the International Monetary Fund (IMF, 2023) estimates that global banking assets exceed USD 400 trillion, with loan portfolios serving as a major source of both profit and financial risk for Institutions. In emerging economies, lending has increased rapidly, with credit to the private sector increasing from about 56% of gross domestic product (GDP) in 2000 to more than 110% in 2022 (World Bank, 2023). While increased lending supports economic growth, it also increases the risk of institutions becoming more exposed to loan repayment problems, especially in economies affected by inflation, economic instability, and weak risk monitoring systems.

Traditional credit risk assessment frameworks have mainly focused on estimating the probability of default (PD), they treat credit risk as a binary classification problem where borrowers are classified as either defaulting or non-defaulting. These models, usually implemented using logistic regression or transition matrix estimation under Markov

assumptions, would give useful but limited views of borrower risk. In reality, credit risk changes over time as borrower financial conditions, repayment behaviour, and general economic conditions gradually change. This process, known as credit risk migration, refers to the movement of borrowers across different credit quality categories, including rating upgrades, downgrades, and eventual default. Research shows that credit deterioration is usually gradual rather than sudden. For example, Standard & Poor's (2023) reports that more than 80% of corporate defaults were first moved into lower credit ratings before defaulting. This shows that it is important to model the different stages of credit deterioration rather than focusing only on final default.

From a Data Science point of view, credit risk migration can be treated as a machine learning problem where borrower information is used to predict future credit risk categories. In other words, the model learns from borrower's financial details, repayment behaviour, and economic conditions to predict the borrower's future credit risk level.

Let $X_t \in \mathbb{R}^n$ denote the feature vector representing borrower-specific financial, behavioral, and macroeconomic attributes observed at time t ,

Let $Y_{t+1} \in \{C_1, C_2, \dots, C_K\}$ represent the borrower's credit state at the subsequent time period.

The objective of predictive credit migration modeling is to learn a function

$$f: \mathbb{R}^n \rightarrow \{C_1, C_2, \dots, C_K\}$$

such that:

$$Y_{\{t+1\}} = f(X_t) + \varepsilon$$

Where ε represents stochastic variability.

This approach helps financial institutions predict borrower credit risk by using patterns found in borrower data. This makes it easier to identify signs of credit problems early and manage risk better.

Traditional credit risk migration models, for example, Markov transition matrices, use past borrower credit movements to estimate future credit changes. Although these models are widely used and easy to understand, they rely on assumptions that may not match real-world borrower behaviour. For example, they often assume that credit risk patterns stay the same over time and do not fully consider differences between borrowers (Lando & Skødeberg, 2002). In reality, borrower behaviour changes due to economic conditions, financial situations, and repayment habits. These weaknesses became more obvious during the global financial crisis, where poor risk models contributed to major credit losses (IMF, 2009).

Recent studies in machine learning have brought in powerful ways to conquer a lot of these limitations. Machine learning models, especially tree-based ensemble methods, can identify complex nonlinear patterns and relationships in large financial datasets more effectively than many traditional models (Breiman, 2001; Lessmann et al., 2015). These models can combine different types of borrower information, including financial ratios, repayment behaviour, and loan characteristics, to give a more detailed understanding of credit risk.

Despite these improvements, most machine learning studies in credit risk still focus mainly on predicting whether a borrower will default or not. Much less research has focused on predicting movements across different credit risk states. This is an important gap because migration-based modelling gives a broader view of how borrower risk changes over time and helps financial institutions detect credit problems earlier. Accurate prediction of credit risk

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changes is also important for portfolio monitoring, regulatory compliance under frameworks such as IFRS 9, and decisions on capital allocation.

This shows there is a clear need for predictive frameworks that can forecast credit risk movements using modern machine learning techniques. This study addresses this gap by treating credit risk outcomes as a multi-class machine learning problem and developing a predictive framework for borrower credit risk outcomes. In doing so, the study combines data-driven machine learning techniques with traditional credit migration concepts to improve credit risk analysis.

Traditional credit risk models mainly relied on transition matrices because earlier datasets contained limited borrower information. Modern datasets provide much richer borrower-level data, but many studies do not fully utilise this information. By combining transition analysis with machine learning, this study provides a more effective approach to understanding and predicting borrower credit risk outcomes.

1.2 Statement of the Problem

Credit risk deterioration remains a major threat to financial institutions and the wider financial system. During the 2008 global financial crisis, banks around the world lost approximately USD 2.8 trillion due to credit-related losses, much of which was caused by borrowers gradually moving into weaker credit positions before eventually defaulting (International Monetary Fund [IMF], 2009). Research also shows that many borrowers usually move into lower credit ratings before eventually defaulting, meaning that credit problems usually happen gradually rather than suddenly (Standard & Poor's, 2023). Since global lending to the non-financial sector

is worth more than USD 160 trillion (Bank for International Settlements [BIS], 2023), being able to detect early signs of credit deterioration is very important for the economy.

Even with this, many traditional credit risk models still focus mainly on predicting the probability of default (PD) as a binary outcome. These models place borrowers into only two groups, default or non-default, and do not properly show how borrower risk changes over time. Traditional migration models, like the Markov transition matrices, estimate the chances of borrowers moving from one credit risk state to another by calculating the probability:

$$P(X_{t+1} = j \mid X_t = i)$$

where the likelihood of moving from state i at time t to state j at time $t + 1$ is derived from historical transition frequencies. Although these models are interpretable, they rely on assumptions that may not match real-world borrower behaviour. For example, they often assume that credit risk patterns stay the same over time and do not fully consider differences between borrowers. Because of this, they may struggle to capture complex financial patterns, borrower behaviour, and changing economic conditions, which reduces their ability to predict credit risk accurately.

Let $X_t \in \mathbb{R}^n$ denote a borrower's feature vector at time t , comprising financial ratios, repayment behaviour, utilisation metrics, and loan characteristics,

let $Y_{t+1} \in \{G_1, G_2, \dots, G_K\}$ represent the borrower's credit grade in a subsequent period. The modelling objective is to learn a predictive function:

$$f: X_t \rightarrow Y_{t+1}$$

that accurately forecasts credit state outcomes and identifies early signs of deterioration.

Many existing credit migration models do not fully use the large amount of borrower-level data available in modern credit datasets to predict changes in credit risk grades. In addition, there is still limited research showing how effective machine learning models are at predicting multiple credit risk outcomes compared to traditional Markov-based models. This creates an important gap in credit risk research, where modern data-driven techniques are still not fully used in migration-based credit modelling.

Therefore, the main problem addressed in this study is the lack of strong predictive models that can accurately analyse credit risk migration using borrower-level data. This study addresses this problem by developing and testing a machine learning-based approach for predicting borrower credit risk outcomes, providing a more detailed and forward-looking understanding of borrower risk behaviour.

1.3 Justification and significance of the study

Global credit markets are becoming larger and more complex, this shows the need for better credit risk prediction models. According to the Bank for International Settlements (BIS, 2023), global credit outside the financial sector exceeds USD 160 trillion, meaning that even small improvements in detecting early credit problems can help reduce financial losses and improve risk management. The importance of this became clear during the 2008 global financial crisis, when banks around the world lost about USD 2.8 trillion due to credit-related losses (International Monetary Fund [IMF], 2009). Many of these losses happened after borrowers gradually moved into weaker credit positions before eventually defaulting. Despite this, many credit risk models still focus mainly on predicting whether a borrower will default or not, making it harder for financial institutions to detect and manage credit problems at earlier stages.

This study is justified because many traditional credit risk models still have important weaknesses. Loan-level datasets contain useful borrower credit information, but traditional migration models rely on past credit movements and do not fully use borrower financial details and repayment behaviour. Because of this, they may struggle to capture complex changes in borrower credit risk. Today, modern credit datasets contain much more borrower-level information, and advances in machine learning create an opportunity to build more accurate and data-driven models for predicting credit risk outcomes

The importance of this study can be explained in three ways. First, from a research perspective, the study helps expand the use of machine learning in credit risk modelling. While many existing studies focus mainly on predicting default, this study focuses on predicting different credit risk outcomes. By treating credit risk migration as a machine learning classification problem, the study adds to current knowledge in credit risk analytics.

Second, from a practical perspective, better prediction of credit risk outcomes can help financial institutions monitor loan portfolios more effectively, identify risky borrowers earlier, and make better risk management decisions. This can improve portfolio monitoring, expected loss estimation, and capital allocation.

Third, from a regulatory perspective, frameworks such as Basel III and IFRS 9 require financial institutions to identify credit problems early and prepare for possible losses in advance. A predictive framework that helps the detection of credit deterioration supports these regulatory goals and strengthens financial system monitoring.

Overall, this study contributes by combining traditional credit migration analysis with machine learning using a real loan-level dataset. Instead of tracking complete borrower credit histories over time, the study uses borrower characteristics, initial credit quality, and final loan outcomes to estimate changes in credit risk behaviour.

1.4 Purpose and Objectives of the Study

The purpose of this study is to develop and evaluate a data-driven framework for predicting credit risk outcomes in loan portfolios. The study treats credit risk outcomes as a machine learning classification problem using borrower-level data. It also checks to see whether machine learning models can improve the prediction of credit risk outcomes compared to traditional migration models such as Markov transition matrices. By using detailed borrower financial and behavioural information, the study wants to improve the early detection of credit deterioration and support better loan portfolio risk monitoring.

Research Objectives

The objectives of this study are to:

1. Construct an empirical credit risk transition matrix benchmark linking initial credit quality and observed loan outcomes.
2. Develop a machine learning model for predicting borrower credit risk outcomes using borrower-level data.
3. Evaluate the performance of the model using classification metrics such as accuracy, precision, recall, and F1-score.
4. Identify the main factors influencing credit risk outcomes using feature importance and SHAP explainability techniques.

5. Examine whether data-driven machine learning approaches provide better insights than traditional migration-based models.
6. Provide practical recommendations for applying machine learning credit risk models in loan portfolio monitoring and risk management.

1.5 Research Questions

This study aims to answer the following questions:

1. How can borrower information be used to predict future credit risk outcomes?
2. How well does an empirical transition matrix describe observed credit risk outcomes?
3. Can machine learning improve the prediction of borrower credit risk outcomes compared to traditional models?
4. Which borrower financial and loan features have the biggest effect on credit risk outcomes?
5. How well can the model identify different stages of credit deterioration?
6. What challenges may arise when using machine learning for credit risk modelling in real financial institutions?

These questions are answered through the methods, analysis, and discussions presented in Chapters 3 to 6. Overall, the study aims to examine how useful machine learning can be for improving credit risk modelling.

1.6 Scope and Limitations of the Study

This study focuses on predicting borrower credit risk outcomes in a consumer loan portfolio using a machine learning approach. Credit risk is represented using different risk categories based on borrower credit sub-grades and loan performance outcomes.

Scope includes:

The study includes:

- Construction of an empirical credit risk transition matrix using borrower credit sub-grades and loan outcomes as a traditional benchmark for migration analysis.
- Development of a Random Forest machine learning model for predicting borrower credit risk outcomes.
- Use of borrower financial and loan characteristics to support predictive modelling.
- Evaluation of model performance using accuracy, precision, recall, and F1-score.
- Use of feature importance and SHAP explainability techniques to identify the main causes of credit risk outcomes.
- Development of practical recommendations for applying machine learning in credit risk monitoring and risk management.

Delimitations

This study has the following boundaries:

- The study only focuses on consumer loan data and does not include corporate or government credit systems.
- Borrower credit sub-grades are used to represent credit risk levels instead of modelling fully observed longitudinal transitions.

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- The study focuses on predicting borrower credit risk outcomes and does not analyse long-term loan survival or lifetime credit losses.
- The study uses only one machine learning model, Random Forest, instead of comparing many different models.
- The analysis only uses borrower financial and loan information available in the dataset.

Limitations

This study also has some limitations:

1. The study uses data from one lending platform, so the results may not fully apply to all financial institutions or credit markets.
2. The dataset does not contain true longitudinal observations of borrower state transitions, requiring a reduced-form approximation of credit risk migration.
3. Economic variables such as inflation and unemployment are not included in the model, which may affect capturing external causes of credit risk.
4. Dataset may contain missing values or exhibit class imbalance that could affect model performance.
5. Historical borrower behaviour may not always predict future credit behaviour accurately, especially during changing economic conditions.

Assumptions

The following were assumed:

1. Borrower credit sub-grades correctly represent ordinal levels of credit risk.
2. Loan outcomes in the dataset correctly reflect borrower credit risk outcomes.
3. Borrower's financial and loan information contains enough useful data for prediction.

4. The Random Forest model can capture important patterns and relationships within the dataset.

1.7 Dissertation Structure

This dissertation is divided into six chapters, with each chapter giving a systematic contribution to the study.

Chapter 1 (Introduction): introduces the background of the study, the research problem, the study objectives, and the scope of the research, establishing the foundation.

Chapter 2 (Literature Review): reviews previous studies and identifies gaps that this study aims to address.

Chapter 3 (Methodology): explains the research design, dataset, feature engineering process, and modelling methods used.

Chapter 4 (Results and Findings): presents the findings and results.

Chapter 5 (Discussion and Implications): explains the findings, discusses practical and theoretical implications, and outlines limitations.

Chapter 6 (Conclusion): summarises the study, and presents final conclusions and suggestions for further research.

CHAPTER 2: Literature Review

2.1 Conceptual Foundations of Credit Risk

Credit risk refers to the possibility that a borrower fails to repay debt according to agreed terms. In financial risk management, credit risk is usually divided into three main parts: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD). These components form the foundation of modern credit risk measurement and are widely used under Basel regulatory frameworks (Basel Committee on Banking Supervision [BCBS], 2006).

PD measures the chance that a borrower will default within a given period. LGD measures the amount of money a lender may lose after default occurs, while EAD refers to the total amount owed when the borrower defaults. Together, these components are used to estimate expected credit loss and support loan pricing, capital allocation, and loan provisioning under frameworks such as IFRS 9 (BCBS, 2006).

Traditional credit risk models mainly focus on predicting whether a borrower will default or not, providing only a limited view of how borrower risk changes over time (Thomas, Edelman, & Crook, 2017).

In reality, borrower credit quality changes gradually over time rather than suddenly. Because of this, financial institutions use internal credit rating systems that place borrowers into different risk categories based on their level of credit quality. These ratings are updated using borrower financial conditions, repayment behaviour, and wider economic conditions, allowing institutions to monitor credit risk better.

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As a result, credit risk is better viewed as a process where borrowers move across different credit risk states over time rather than moving directly from performing to default. In this framework, default is seen as the final stage of gradual credit deterioration. Modelling these movements, known as credit risk migration, provides a better understanding of borrower risk because it helps identify early signs of credit deterioration that may not appear in simple default prediction models.

Understanding credit risk migration is therefore important for loan portfolio monitoring, early warning systems, and forward-looking risk management. By tracking changes across different credit risk states, financial institutions can identify potential problems earlier, improve risk prediction, and take action before borrowers eventually default..

2.2 Credit Risk Migration and Rating Transition Dynamics

Credit risk migration refers to the movement of borrowers across different credit risk levels over time. These risk levels are usually based on internal credit ratings used by financial institutions or external ratings provided by agencies such as Standard & Poor's and Moody's. These rating systems help order borrowers according to their level of credit risk.

Credit ratings place borrowers into different risk categories, ranging from low-risk borrowers to borrowers facing financial problems. These ratings can change over time depending on borrower financial conditions, repayment behaviour, and economic conditions. As a result, borrowers may move into higher-risk categories, lower-risk categories, or remain in the same category.

Credit migration can be represented as a stochastic state transition process.

Let S_t denote the credit state of a borrower at time t , where $S_t \in \{1, 2, \dots, K\}$ represents a finite set of ordered credit risk states.

Credit risk migration can also be viewed as a process where borrowers move from one credit state to another over time. The chances of moving between these credit states are called transition probabilities, and they form the foundation of migration-based credit risk models (Jarrow, Lando, & Turnbull, 1997).

Compared to normal default prediction models, credit risk migration models provide a broader understanding of borrower risk. Because of this, credit migration analysis plays an important role in loan portfolio management. Financial institutions use migration analysis to monitor portfolio quality, track changes in borrower credit risk and estimate effects on losses, capital requirements, and loan provisions. Studies have also shown that borrower downgrades become more common during periods of economic stress (Nickell, Perraudin, & Varotto, 2000).

2.3 Classical Credit Migration Modelling Approaches

One of the most common methods used to model credit risk migration is the Markov transition matrix model. This model estimates the chances (transition probability matrix) of borrowers moving from one credit risk category to another over time.

Let S_t denote the credit state at time t .

The probability that a borrower transitions from state i to state j between two time periods is given by:

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$$p_{ij} = P(S_{t+1} = j \mid S_t = i)$$

The resulting matrix satisfies the stochastic property:

$$\sum_{j=1}^K p_{ij} = 1 \forall i$$

In this approach, the resulting **transition matrix** is P , borrowers are placed into different credit risk states, and P are used to estimate the likelihood of moving between these states. Therefore P contains the probabilities of all possible credit risk movements. Each row of the matrix adds up to one because every borrower must move to a credit state in the next period.

Markov transition models became widely used after the development of the *CreditMetrics* framework by Gupton, Finger, and Bhatia (1997). The framework showed how transition matrices could be used to estimate portfolio credit losses and measure financial risk. Similarly, Jarrow, Lando, and Turnbull (1997) developed another framework linking credit spreads to credit rating movements, further increasing the importance of migration-based credit risk analysis.

Many studies have also examined how borrower credit ratings change over time. For example, Nickell, Perraudin, and Varotto (2000) found that transition probabilities change during different economic conditions, especially during financial crises and economic stress. This shows an important weakness of standard Markov models, which often assume that credit risk patterns remain constant over time.

Although Markov transition models are widely used, other credit risk models also exist. Structural models, introduced by Merton (1974), explain default as when a company's assets

fall below its debt obligations. Reduced-form models treat default as a random event influenced by economic conditions and other observable factors (Jarrow et al., 1997).

In practice, transition matrices are usually created using historical borrower rating data collected by credit rating agencies or financial institutions. Agencies such as Moody's and Standard & Poor's publish long-term transition matrices that are widely used as benchmarks for credit migration analysis (Nickell et al., 2000).

Overall, Markov transition matrix models are one of the most widely used approaches for analysing credit risk migration because they are simple, easy to understand, and useful for portfolio risk analysis. However, their simplifying assumptions beg for more flexible models that can better capture borrower-level differences and complex credit risk patterns.

2.4 Limitations of Classical Credit Migration Models

Although Markov transition matrix models are widely used in credit risk analysis, they have several important limitations.

First, Markov models assume that future borrower risk depends only on the borrower's current credit state. In reality, borrower risk is also affected by past repayment behaviour, previous downgrades, and earlier financial problems. Because of this, Markov models may not fully capture how borrower risk changes over time.

Secondly, traditional transition models often assume **time-homogeneous transition probabilities**, meaning that credit risk patterns remain the same over time. However, studies show that borrower credit risk changes during different economic conditions, especially during

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financial crises and economic downturns. This means that fixed transition matrices may not reflect real-world credit behaviour accurately.

Thirdly, these models treat borrowers within the same credit category as if they are all similar. In reality, borrowers differ in income, loan structure, financial condition, and repayment behaviour. As a result, traditional models may fail to use important borrower-level information available in modern credit datasets.

Fourthly, transition probabilities are estimated using historical borrower movements, this can create problems when transition events are sparse or unevenly distributed. Especially in credit datasets exhibiting class imbalance, where high-risk or default states are underrepresented, the estimated transition probabilities may become less reliable.

Finally, and most critically in practical applications, traditional migration models usually require complete borrower credit histories over time. However, many real-world loan datasets do not contain full longitudinal borrower information. This makes it difficult to apply traditional Markov-based models directly and hence necessitates alternative modelling approaches that can operate under reduced-form data conditions.

Overall, these limitations show that traditional migration models may struggle to capture the nonlinear, heterogeneous, and data-intensive nature of modern credit risk environments. As financial datasets become larger and more detailed, there is a growing need for more flexible and data-driven approaches.

Machine learning models provide one possible solution because they can use large amounts of borrower-level information to predict credit risk outcomes more flexibly, even when full borrower transition histories are not available.

2.5 Statistical Approaches to Multi-Class Credit Modelling

Apart from transition matrix models, statistical models have also been used to predict different credit risk outcomes. These models extend traditional default prediction methods by allowing borrowers to be classified into several credit risk categories instead of only default and non-default states.

In this approach, borrower financial and loan information is used to predict different credit risk categories. This is better in contexts where credit quality is represented through ordered rating categories or derived risk states. This provides a more detailed understanding of borrower risk compared to simple default prediction models.

One of the most common statistical methods used for this purpose is (Multinomial logistic regression)MLR, which extends binary logistic regression to problems involving multiple outcome categories.

Let $Y \in \{1, 2, \dots, K\}$ denote the credit risk state. The probability of observing outcome k is defined as:

$$P(Y = k | X) = \frac{\exp(\beta_k X)}{\sum_{j=1}^K \exp(\beta_j X)}$$

MLR is popular because it is easy to interpret. The model helps explain how borrower characteristics influence different credit risk outcomes, which is important in financial institutions where models must be transparent and understandable.

Other statistical models, such as ordered logistic and ordered probit models, are also used when credit risk categories follow a natural ranking. However, these models rely on assumptions that may not always work well in real-world credit datasets.

Despite their usefulness, statistical models also have important limitations. They assume simple linear relationships between borrower variables and credit risk outcomes, making it difficult to capture complex borrower behaviour. Their performance may also reduce when working with large datasets or datasets with class imbalance, where low risk states are underrepresented.

Because of these limitations, statistical models are often used as benchmark models rather than the best predictive tools for modern credit risk analysis. When modelling credit risk outcomes using large-scale, non-longitudinal loan datasets, more flexible methods are required to capture nonlinearities and heterogeneous borrower behaviour.

For this reason, although statistical models provide an important foundation for credit risk classification, this study adopts a machine learning approach to improve the prediction of borrower credit risk outcomes under realistic data conditions.

2.6 Machine Learning Approaches to Credit Risk Prediction

Improvements in data collection and computing power have increased the use of machine learning (ML) in credit risk modelling. Unlike traditional statistical models, machine learning models learn patterns directly from data, making it easier to capture complex borrower behaviour and relationships within large financial datasets. Many studies have shown that machine learning models often perform better than traditional statistical models, especially when working with large and complex credit datasets.

Among ML techniques, a tree-based ensemble method is the Random Forest. Random Forest combines many decision trees to improve prediction accuracy and reduce errors. This makes it useful for credit datasets characterised by heterogeneity, noise, and complex borrower interactions.

Machine learning models can use different types of borrower information and capture complex relationships; these properties are important in credit risk analysis.

However, ML models also have some challenges. Some models are difficult to interpret and may overfit the data if not properly validated. Because of this, proper validation methods and explainability techniques are important to ensure reliable predictions.

Worthy to note, in practical settings where longitudinal credit state transitions are not observed, machine learning models can still predict borrower credit risk outcomes using available borrower-level information.

Recent studies also show that ML models perform well in credit risk analysis. Bussmann et al. (2021) show that tree-based ensemble methods often perform better than traditional statistical models in large credit datasets. Similarly, Khandani et al. (2010) showed that using detailed borrower information improves the prediction of credit risk outcomes. These studies highlight the growing importance of machine learning in modern credit risk analysis.

More recent studies have also shown the growing use of advanced machine learning methods in financial risk analysis. For example, Sirignano et al. (2018) showed that large-scale machine learning models can improve mortgage default prediction, while Bellotti and Crook (2009) found that nonlinear models improve credit scoring performance. These studies further highlight the growing role of machine learning in modern credit risk modelling.

For this reason, this study uses a Random Forest model to predict borrower credit risk outcomes using loan-level data. Random Forest was selected for its strong predictive performance, handles data irregularities very well, and works well for modelling complex, nonlinear relationships within non-longitudinal credit datasets

2.7 Feature Engineering in Loan-Level Credit Data

Feature engineering is an important part of credit risk modelling because the quality of the input data affects how well the model performs. In loan-level datasets, borrower financial details, loan information, and repayment behaviour are converted into useful features that help predict credit risk outcomes.

Borrower financial variables are some of the most important features because they show a borrower's ability to repay loans. For example, annual income, debt-to-income ratio, and revolving credit balance, they help explain borrower financial condition and credit risk.

Behavioural variables are also important in credit risk prediction. Features such as repayment history and delinquency records help identify early signs of financial problems and make it easier to separate low-risk borrowers from high-risk borrowers.

Loan-related variables also improve model performance by showing differences in loan structure and borrower risk. Examples include loan amount, interest rate, loan purpose, and borrower credit grade.

The dataset used in this study does not contain complete longitudinal borrower histories. Because of this, feature engineering is performed within a cross-sectional framework, where borrower characteristics at observation are used to predict future credit risk outcomes. This limits the use of temporal features and instead focuses on static and behavioural indicators as representations of underlying credit dynamics.

Overall, feature engineering helps organise borrower data into a structured format that allows machine learning models to identify patterns between borrower characteristics and credit risk outcomes. By focusing on meaningful borrower and loan features, the study improves both prediction performance and model interpretability.

2.8 Explainable AI in Credit Risk Modelling

Although ML models can predict credit risk very well, they are often difficult to understand. Models such as Random Forest can identify complex non-linear patterns in borrower data, but it is not always easy to explain how they make decisions. This creates problems in financial institutions where models must be transparent and explainable.

Explainable Artificial Intelligence (XAI) helps solve this problem by providing methods that explain model predictions and show how different features affect the results. This is important in credit risk modelling because financial institutions and regulators need to understand why a model makes certain predictions.

One of the most widely used explainability methods is Shapley Additive Explanations (SHAP). SHAP measures how much each feature contributes to a model's prediction and helps explain how the model works.

SHAP provides both global and local explanations. Global explanations show the most important features across the entire dataset, while local explanations show how borrower characteristics affect individual predictions. This is useful in credit risk analysis because financial institutions need both portfolio-level and borrower-level insights.

As ML models become increasingly integrated into financial decision-making, explainability has become a major research focus. Recent studies have highlighted SHAP as one of the most effective approaches for interpreting credit scoring models while maintaining predictive performance (Arrieta et al., 2022).

The importance of explainability is also supported by financial regulations, which require credit risk models to be transparent, understandable, and properly validated. Molnar (2020) explains that interpretability is important for building trust, accountability, and regulatory compliance in machine learning models. This is especially important in finance, where model predictions must be both accurate and easy to justify.

2.9 Ethical and Regulatory Considerations

Using machine learning in credit risk modelling raises important ethical and regulatory concerns because these models can affect lending decisions and access to loans. Because of this, models must be fair, transparent, and responsibly developed.

One major concern is **algorithmic bias**. Machine learning models learn from historical credit data, and this data may contain existing social or economic biases. If not properly managed, the model may make unfair predictions for some borrower groups. To reduce this risk, careful feature selection, data preprocessing, and model evaluation are important.

Data privacy is also very important. Credit datasets usually contain detailed borrower information. However, this study uses a public and anonymized dataset, meaning that no personal borrower identities are included.

From a regulatory perspective, credit risk models must meet standards related to **model validation, governance, and transparency** under frameworks such as Basel III and IFRS 9. Financial institutions are expected to ensure that models are reliable, properly tested, and well documented. Recent regulatory discussions have emphasized the need for transparent and explainable machine learning systems within financial services. Because of this, **model**
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interpretability is very important. Explainability techniques such as SHAP help improve transparency by showing how different features influence model predictions.

Another important issue is **model risk**, which refers to the possibility that models may produce incorrect or unstable predictions due to data problems or model limitations. In this study, possible sources of model risk include data limitations, class imbalance, and the lack of complete borrower credit histories over time. To reduce these risks, proper preprocessing and validation methods are applied to ensure reliable model performance.

Overall, ethical and regulatory considerations are important in credit risk modelling to ensure that machine learning models remain fair, transparent, reliable, and suitable for responsible financial decision-making.

2.10 Comparative Synthesis of Related Studies

Many studies have examined credit risk modelling using traditional statistical models, transition matrix models, and machine learning approaches. These studies differ in the type of data used, modelling methods, and prediction goals. Comparing these studies helps show how credit risk modelling has developed over time and explains where this study fits within existing research.

Study	Dataset Context	Modelling Approach	Key Methods	Key Findings
Jarrow, Lando & Turnbull (1997)	Corporate bond ratings	Classical migration modelling	Markov transition models	Introduced transition matrices for modelling credit risk changes
Nickell, Perraudin & Varotto (2000)	Corporate credit ratings	Empirical migration analysis	Transition matrices	Found that credit risk changes during different economic conditions

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Study	Dataset Context	Modelling Approach	Key Methods	Key Findings
Lessmann et al. (2015)	Benchmark credit datasets	Machine learning comparison	Random Forest, SVM, neural networks	Showed that machine learning models perform better than statistical models
Khandani, Kim & Lo (2010)	Consumer credit portfolios	Data-driven modelling	Machine learning techniques	Showed the importance of borrower-level data in credit prediction

Table 2.1: Comparative Summary of Selected Credit Risk Modelling Studies

Analytical Discussion

It is seen that credit risk modelling has changed significantly over time. Earlier studies mainly focused on Markov transition models, which were used to analyse borrower movements across credit risk categories. These were easy to interpret but depended on restrictive assumptions and longitudinal data.

Later, statistical models such as logistic regression (LR) became widely used for default prediction. These models are easy to explain and work well in regulated financial environments, but they mainly focus on simple linear relationships and binary outcomes.

Recent advances in machine learning have expanded the application of explainable AI, ensemble learning, and regulatory-compliant predictive modelling in credit risk management. Studies published between 2021 and 2025 increasingly emphasize model transparency, fairness, and alignment with regulatory frameworks such as IFRS 9. These developments suggest a growing shift towards combining predictive accuracy with explainability and governance requirements.

However, two important gaps still exist in the literature.

First, this narrow focus on purely default prediction often overlooks early signs of credit deterioration.

Second, the reliance on **fully observed longitudinal rating data**, which is not always the case in real-world datasets, greatly limits research on modelling credit risk movements using incomplete or non-longitudinal datasets.

This study addresses these gaps by using a machine learning framework to predict multiple credit risk outcomes using non-longitudinal loan-level data. By combining an empirical transition matrix with a Random Forest machine learning model, the study shows how machine learning can be used to model credit risk migration in realistic datasets where traditional assumptions may not fully apply.

2.11 Research Gaps and Summary

2.11.1 Research Gaps

Although credit risk modelling has been widely studied, some important gaps still remain. Many traditional approaches depend on borrower credit histories over time, which are often unavailable in real-world datasets. At the same time, many prediction models focus mainly on default and may not provide a broader view of borrower risk.

Machine learning has shown strong potential in credit risk prediction, but there is still limited research on how it can be applied to non-longitudinal loan data while remaining transparent and easy to interpret.

This study addresses these gaps by using machine learning to predict borrower credit risk outcomes from non-longitudinal data. It also combines predictive modelling with explainable AI to improve both prediction and model transparency

Taken together, these gaps show the need for a modelling approach that:

- predicts multiple credit risk outcomes,
- works well even with complex data structures,
- balances prediction accuracy with model interpretability.

2.11.2 Contribution of the Present Study

This study addresses these gaps by developing a machine learning framework for predicting borrower credit risk outcomes across multiple credit states using loan-level data.

This study uses a reduced-form approach, where borrower characteristics predict future credit risk outcomes. This provides a practical way to model credit risk changes when full borrower transition data is not available.

The study makes three main contributions:

1. Methodological Contribution

The study treats credit risk modelling as a multi-class prediction problem using non-longitudinal data, helping connect traditional migration models with modern machine learning approaches.

2. Empirical Contribution

The study uses a Random Forest model to capture complex relationships between

borrower characteristics and credit risk outcomes, showing how machine learning can improve credit risk prediction.

3. Interpretability Contribution

The study uses SHAP explainability techniques to identify the most important factors affecting credit risk predictions, improving model transparency and supporting regulatory requirements.

Overall, the study shows how machine learning can be used to predict borrower credit risk outcomes in real-world loan datasets, giving a more flexible and practical alternative to traditional credit risk models.

2.11.3 Chapter Summary

This chapter reviewed important ideas and studies related to credit risk modelling, especially credit risk migration and predictive modelling methods. Traditional models, were discussed, as well as their limitations in explaining complex borrower credit behaviour.

The chapter also examined the growing use of machine learning in credit risk analysis, showing how it can capture complex relationships in large borrower datasets while also raising important issues related to explainability and model governance.

Finally, the chapter identified important gaps in existing research and explained how this study uses a data-driven machine learning approach to model multiple credit risk outcomes under realistic data conditions. These discussions provide a good foundation for the methodology presented in the next chapter.

CHAPTER 3: Research Methodology

3.1 Introduction

This chapter explains the methods used to develop a predictive model for borrower credit risk outcomes using loan-level data. Based on the ideas and research gaps discussed in Chapter 2. The research design, dataset, and preprocessing methods used to prepare the modelling data are explained. It also discusses the feature engineering process and the development of the Random Forest model.

To provide a traditional benchmark, the study also constructs an empirical transition matrix using observed borrower credit states and loan outcomes. Model performance is evaluated using multi-class classification metrics, while SHAP explainability techniques are used to identify the main factors influencing credit risk predictions.

Overall, this chapter presents a practical and data-driven framework for modelling credit risk when complete borrower transition histories are not available.

3.2 Research Design

This study uses a quantitative and data-driven research design to analyse borrower credit risk outcomes using loan-level data. The study treats credit risk prediction as a supervised multi-class classification problem, where borrower information is used to predict different credit risk categories.

The study follows an empirical and non-experimental design using a large secondary loan-level dataset. The analysis focuses on identifying patterns in borrower financial information, repayment behaviour, and loan characteristics.

The modelling process follows these main steps:

1. **Data Acquisition:** Collection of a secondary loan-level dataset containing borrower financial and loan performance information.
2. **Data Preprocessing:** Cleaning and preparing the dataset by handling missing values, encoding categorical variables, and addressing class imbalance.
3. **Feature Engineering:** Creation of meaningful borrower and loan features representing financial condition, repayment behaviour, and loan characteristics.
4. **Model Development:** Development of a Random Forest model to predict borrower credit risk outcomes.
5. **Model Evaluation:** Evaluation of model performance using metrics such as accuracy, precision, recall, and F1-score.
6. **Model Interpretation:** Use of SHAP explainability techniques to identify the most important factors influencing model predictions and improve model transparency.
7. **Empirical Transition Matrix Construction:** Construction of an empirical transition matrix to provide a traditional benchmark for analysing borrower credit risk movements using observed loan outcomes.

Overall, the research design provides a practical and structured framework for modelling borrower credit risk using realistic loan-level data.

3.3 Data Sources and Dataset Description

3.3.1 Data Sources

The success of any credit risk model depends on the quality of data used. This study uses a secondary dataset from the Lending Club Loan Data repository available on Kaggle (Lending Club, 2018). The dataset contains anonymised records of consumer loans issued between 2007 and 2018, including borrower financial information, loan details, and loan outcomes.

The Lending Club dataset was selected because it contains a large amount of borrower-level information and is widely used in credit risk research. It provides useful information for analysing borrower behaviour and predicting credit risk outcomes.

The dataset contains three main types of variables:

- **Borrower financial characteristics**
(e.g., annual income, debt-to-income ratio, credit history information)
- **Loan attributes**
(e.g., loan amount, interest rate, loan grade, loan purpose)
- **Behavioural and performance indicators**
(e.g., delinquency history, repayment status, loan outcome)

For this study, credit risk states are created from loan performance outcomes. This allows borrowers to be grouped into different credit risk categories instead of simply classifying them as default or non-default.

Although the dataset comes from a peer-to-peer lending platform rather than a traditional bank, it still provides detailed borrower information that makes it suitable for credit risk analysis. Many previous studies have also used Lending Club data for machine learning-based credit risk research.

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The dataset is publicly available and anonymised, meaning that no personal borrower information is included. All analysis is conducted for academic purposes and follows ethical research standards.

3.3.2 Dataset Summary/Structure
Dataset Characteristic Description

Data source	Lending Club Loan Data (Kaggle)
Observation unit	Individual borrower loan record
Time period	2007 – 2018
Dataset type	Secondary structured financial dataset
Number of observations	Approximately 2 million loan records
Number of variables	More than 100 borrower and loan attributes
Feature categories	Financial, loan, and behavioral variables
Target variable	Derived credit risk state
Prediction objective	Predict borrower credit risk outcome

Table 3.1: Summary of Dataset Characteristics

The dataset contains a large number of variables, so feature selection is needed to identify the most useful variables for modelling. Variables are selected based on their importance in credit risk analysis and data availability. The focus is mainly on borrower financial information, loan characteristics, and behavioural indicators.

The target variable is created from the *loan_status* field, which shows the final outcome of each loan. These outcomes are grouped into different credit risk categories to support a multi-class classification framework.

Key Variables Used in the Study

The final dataset includes the following important variables:

- **Loan_ID** — Unique number for every loan taken.
- **Loan_Amount** — Total amount borrowed by the borrower
- **Interest_Rate** — Interest rate charged on the loan
- **Annual_Income** — Borrower's yearly income
- **Debt_to_Income_Ratio (DTI)** — Measure of borrower debt level
- **Employment_Length** — Number of years the borrower has worked
- **Loan_Purpose** — Reason for taking the loan
- **Loan_Grade** — Credit grade assigned to the borrower
- **Delinquency_History** — Number of formerly missed payments
- **Revolving_Balance** — Outstanding revolving credit balance
- **Loan_Status** — Final outcome of the loan
- **Credit_Risk_State** — Credit risk category used as the target variable

3.3.3 Construction of Credit Risk States

To predict borrower credit risk, the original Lending Club loan data was converted into different credit risk categories. These categories are used as the target variable in the machine learning model.

The dataset contains two important indicators of credit risk:

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- **Loan status**, which shows the final outcome of a loan (e.g., Fully Paid, Current, Charged-Off, Default)
- **Sub-grade classification**, shows the borrower's credit quality when the loan was first issued (A1 to G5).

In this study, credit risk states are created using **loan status**, while **sub-grade** is used as an input feature for prediction. To support multi-class credit risk modelling, the original loan outcomes were grouped into three broader credit risk categories

Construction of Credit Risk Outcome Classes

Outcome State	Credit Risk Outcome Class
Paid	Low Risk Outcome
Active	Intermediate outcome
Default	Adverse Outcome

Table 3.2: Construction of Credit Risk Outcome Classes

The resulting variable, named *migration_class* in the implementation stage, serves as the target variable for the model development. Although referred to as a migration class, the categories represent observed credit risk outcomes rather than directly observed borrower transitions over time.

Interpretation of Credit Risk Outcome Classes

Credit Risk Outcome Class	Meaning
Low Risk Outcome	Borrowers making payments as expected

Credit Risk Outcome Class	Meaning
Intermediate Outcome	Borrowers showing early repayment problems
Adverse Outcome	Borrowers who defaulted, were charged off, or experienced severe repayment difficulties

Table 3.3: Interpretation of Credit Risk Outcome classes.

Methodological Justification

Using multiple credit risk states provides several benefits:

- It captures different levels of borrower risk instead of focusing only on default.
- It provides more detailed information about borrower credit quality.
- It supports early identification of borrowers whose credit risk is increasing.
- It allows the use of multi-class machine learning models to predict different credit risk outcomes.
- It aligns with the objective of developing a reduced-form framework for analysing borrower credit risk outcomes.

Reduced-Form Modelling Consideration

It is important to say that the Lending Club dataset does not contain complete borrower credit histories over time. Because of this, the study cannot directly track how borrowers move between different credit risk states.

Instead, borrower characteristics are used to predict the likelihood of belonging to a particular credit risk category. This provides a practical, data-driven way of estimating credit risk changes using the information available in the dataset.

3.3.4 Formal Modelling Framework

The main goal of the model is to predict a borrower's credit risk category using borrower information. Due to the structure of the dataset, the problem is going to be treated as a supervised multi-class classification task.

In this study, borrower financial information, repayment behaviour, and loan characteristics are used as input variables, while the credit risk category is the outcome being predicted. The model estimates the probability that a borrower belongs to a particular credit risk category based on their characteristics.

Let X denote the feature.

Let $Y \in \{1, 2, \dots, K\}$ denote the corresponding credit risk category, where K represents the number of discrete risk classes.

The modelling task is therefore defined as estimating the conditional probability:

$$P(Y = k | X)$$

where:

- X represents borrower characteristics
- Y represents the observed credit risk category
- k represents one of the possible credit risk states.

Unlike traditional credit migration models, this approach focuses on the relationship between borrower characteristics and observed credit risk outcomes. This allows the model to predict different levels of borrower risk rather than simply predicting whether a borrower will default or not.

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Formally, let $X_i = (x_{i1}, x_{i2}, \dots, x_{ip})$ denote the vector of borrower-level features for loan observation i , including financial, behavioural, and loan-specific variables.

Let $Y_i \in \{1, 2, \dots, K\}$ denote the derived credit risk outcome class, where K represents the number of outcome categories.

The modelling objective is therefore to estimate:

$$P(Y_i = k | X_i), k = 1, 2, \dots, K$$

Unlike a fully longitudinal migration framework, this study does not estimate $P(S_{t+1} = j | S_t = i)$ from repeated borrower state observations. Instead, the model then estimates the probability that a borrower belongs to each risk category.

3.3.5 Implication for Model Design

Creating the target variable from credit risk categories means that the problem can be treated as a multi-class classification task, where borrowers are grouped into different risk levels.

This has several advantages:

- It allows the use of machine learning models such as Random Forest to capture complex patterns in borrower data.
- It can handle differences in borrower behaviour better than traditional linear models.
- It works well even when complete borrower credit histories are not available.
- It predicts multiple levels of credit risk instead of only default and non-default outcomes.

Because of this, the model provides a practical and data-driven way of analysing borrower credit risk while also remaining consistent with the structure and limitations of the dataset.

3.4 Data Preprocessing

Before building the model, the dataset was cleaned and prepared to improve data quality and model performance. Financial datasets often contain missing values, inconsistent formats, and extreme values also known as outliers, which can affect prediction results if they are not properly handled.

The preprocessing process included data cleaning, handling missing values, treating outliers, encoding categorical variables, correcting class imbalance, and preparing the final dataset for modelling.

3.4.1 Data Cleaning

The first step was to check the dataset for duplicate records, invalid values, and data inconsistencies. Duplicates were removed so that each record represented was unique.

Variables with a large number of missing values or little relevance to the study were removed.

This helped improve data quality and reduce unnecessary information in the dataset.

Handling Missing Values

Missing values were handled using different methods depending on the type of variable:

- **Numerical variables**, such as annual income and debt-to-income ratio, were filled using the **median** value.

- **Categorical variables** were filled using the **mode** or assigned a separate “missing” category where needed.

These actions help retain dataset size while maintaining the overall structure of the dataset.

Outlier Treatment

Some financial variables, such as loan amount, annual income, and debt-to-income ratio, contained extremely large values that could affect model performance.

To reduce their impact, these variables were capped at the 99.5th percentile. This allowed the study to keep important high-risk observations while preventing extreme values from having too much influence on the model.

Categorical Variable Encoding

Some variables, such as loan purpose, employment length, and loan grade, are categorical and cannot be used directly by machine learning models.

To solve this, categorical variables were converted into numerical form using encoding techniques.

Nominal variables were transformed using one-hot encoding, which creates separate binary variables for each category. This allows the model to learn the effect of each category without assuming any ranking between them.

Feature Scaling

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All numerical variables were converted into the correct numeric format before model training. Data type checks were also performed to ensure that all variables were suitable for analysis and to prevent errors during modelling.

Class Imbalance Handling

The dataset contains more low-risk loans than high-risk loans. This creates a problem called class imbalance, where the model may focus too much on the larger groups and perform poorly when predicting high-risk borrowers.

He and Garcia (2009) highlight that class imbalance is a fundamental challenge in classification tasks and requires appropriate handling to ensure reliable model performance. In this study, to reduce this problem, **class weighting** was used during model training. This gives more importance to smaller risk categories and helps the model identify high-risk borrowers better.

Final Dataset Preparation

After preprocessing, the dataset was checked to ensure that it was complete, consistent, and ready for modelling. To improve representation, samples were selected from each year between 2007 and 2018.

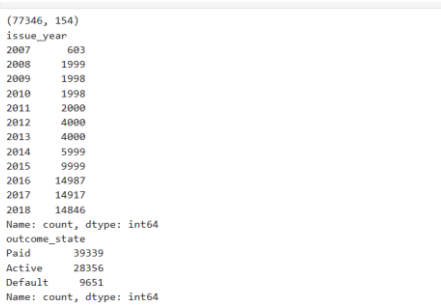


Figure 3.1 Sample Distribution Across Origination Years

The final dataset was divided into:

- **Feature matrix (X):** borrower financial, behavioural, and loan characteristics
- **Target variable (Y):** The borrower's credit risk category

This process ensured that the final dataset was clean, organised, and suitable for model development.

3.5 Feature Engineering Strategy

Feature engineering is an important part of predictive modelling because it converts raw borrower data into useful variables that help the model make better predictions.

In this study, feature engineering focused on creating meaningful variables that explain changes in borrower credit risk across different credit risk categories.

Formally, the model uses borrower features (**X**) to predict the borrower's credit risk category (**Y**):

$$X \in \mathbb{R}^{n \times p}, y \in \{1,2,3,4,5\}$$

where:

- X represents the matrix of engineered borrower-level features
- y represents the discrete credit risk state
- n is the number of observations and p is the number of features

3.5.1 Feature Construction

The original dataset variables were grouped into three main categories.

(i) Financial Capacity Features

These features measure a borrower's ability to repay loans and handle financial pressure.

They help explain why some borrowers move into higher-risk categories.

The features include:

- **Debt-to-Income Ratio (DTI):** Measures how much debt a borrower has compared to income.

$$DTI = \frac{\text{Total Monthly Debt Payments}}{\text{Monthly Income}}$$

- **Log-transformed Annual Income:** Used to reduce extreme income values and improve data distribution.

$$\log(\text{Income})$$

- **Credit Utilisation Proxy:** Measures how much available credit a borrower is using.

$$\frac{\text{Revolving Balance}}{\text{Total Credit Limit (proxy)}}$$

- **Credit History Length:** Measures how long the borrower has been using credit.

These variables help describe borrower financial strength and credit risk.

(ii) Loan Characteristics

These features describe the loan itself and help explain differences in borrower risk.

The features include:

- **Loan Amount:** Total amount borrowed
- **Interest Rate:** Interest charged on the loan.
- **Loan Term:** Whether the loan lasts for 36 months or 60 months.
- **Loan Grade:** Credit grade assigned to the borrower (A–G → 1–7).

These variables help capture the level of risk associated with each loan.

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(iii) Behavioral Credit Features

Behavioural features show how borrowers have managed credit in the past and can provide early signs of financial problems.

The features include:

- **Delinquency Count:** Number of previous missed or late payments.
- **Open Credit Lines:** Number of active credit accounts held by the borrower.
- **Payment Behaviour Indicators:** Information based on the borrower's loan repayment status.

These features are important because borrowers often show signs of financial difficulty before they eventually default.

3.5.2 Feature Selection Strategy

Features were selected based on both financial importance and data quality.

Variables were included if they:

- Are commonly used in credit risk analysis.
- Contained reliable and complete data.
- Improved model performance during training.

Variables that provided similar information or added little value were removed to reduce unnecessary complexity and improve performance. Also, variables containing post-loan information were excluded from model training. These variables become available only after loan issuance or during the repayment period and would therefore not be known at the time a lending decision is made. Removing such variables helped prevent data leakage and ensured that model performance reflected information that would realistically be available when assessing borrower credit risk.

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3.5.3 Link to Credit Risk Migration

The engineered feature set help the model to learn:

$$P(S_{t+1} = j \mid X_t)$$

where borrower characteristics at time t are used to predict transitions into future credit risk states j .

In simple terms, the model uses borrower information available today to predict the borrower's future credit risk category.

This approach supports the study's goal of modelling credit risk changes across different risk levels rather than simply predicting default or non-default outcomes.

3.6 Model Development and Training

This study develops a predictive model to estimate borrower credit risk outcomes using a supervised multi-class classification approach. The goal is to learn patterns between borrower characteristics and observed credit risk categories.

3.6.1 Empirical Transition Matrix Benchmark

To provide a traditional benchmark, an empirical transition matrix was created using the observed loan data.

The transition matrix shows the probability of borrowers moving from an initial credit quality category to a final loan outcome category. This provides a simple way to understand credit risk patterns within the dataset.

Formally, the transition probability from state i to state j is defined as:

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$$P_{ij} = \frac{N_{ij}}{\sum_j N_{ij}}$$

where N_{ij} represents the number of observations transitioning from initial state i to outcome state j . In the resulting matrix, each row sums to one, showing conditional transition probabilities.

The matrix was created using borrowers' initial sub-grade classifications and their final loan outcomes.

To overcome the absence of longitudinal credit history data, this study uses the relationship between borrowers' initial credit quality and final loan outcomes as an approximation of credit risk outcomes.

Although this is not a full Markov migration model, it still provides useful information about how borrower risk changes and serves as a benchmark for comparing the machine learning model.

The models were trained using past borrower observations, where borrower characteristics were used as input variables and the derived credit risk category was used as the target variable.

Although other modelling approaches could have been used and then compared, this study focuses on building a reliable and interpretable predictive framework rather than comparing many different models. This allows for a deeper understanding of model performance and the key factors influencing borrower credit risk outcomes.

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3.6.2 Random Forest Approach

This study uses **Random Forest** as the main machine learning model. Random Forest was selected because it performs well, handles complex data effectively, and is less likely to overfit compared to many other machine learning models.

The choice of Random Forest is also due to its performance large financial datasets that contain many borrower and loan variables. Biau and Scornet (2016) found that the model performs reliably in high-dimensional datasets, while Probst et al. (2019) showed that it remains stable across different prediction tasks. The model can identify complex relationships between borrower affordability, loan characteristics, repayment behaviour, and credit quality without requiring strict statistical assumptions.

Model Specification

Random Forest works by creating many decision trees and combining their predictions to produce a final result (Breiman, 2001). Instead of relying on a single decision tree, the model combines the predictions of many trees, making the results more accurate and reliable.

The Random Forest prediction function can be expressed as:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(X)$$

where $f_t(X)$ represents the prediction of the t^{th} decision tree and T denotes the total number of trees in the ensemble.

This approach reduces errors and improves performance, especially when working with large datasets that contain many variables (Lessmann et al., 2015).

Feature Set and Target Variable

The model was trained using features that represent:

- borrower financial capacity
- loan characteristics
- Borrower repayment behaviours

The target variable is the **credit risk category**, which was created from the loan status variable.

3.6.3 Training Procedure

The Random Forest model was developed in Python using the **scikit-learn** library. To make the evaluation more realistic and to preserve chronological ordering within the dataset, a **temporal train-test split** was used. Loans issued up to 2015 were used to train the model, while loans issued after 2015 were used to test the model.

Although a random 80:20 split is commonly used in machine learning studies, a temporal split was chosen because it better reflects real-world credit risk prediction. In practice, financial institutions use past borrower data to predict future outcomes, not the other way around.

Since the dataset contains more low-risk borrowers than high-risk borrowers, **class weighting** was applied during training. This gives more importance to high-risk borrowers and helps the model identify them more effectively.

3.6.4 Hyperparameter Optimisation and Cross-Validation

To improve model robustness and reduce the risk of overfitting, hyperparameter optimisation was performed using GridSearchCV.

A 5-fold cross-validation strategy was adopted, whereby the training dataset was partitioned into five equally sized folds. During each iteration, four folds were used for model training while the remaining fold was used for validation. This process was repeated five times so that each fold served as a validation set once. Cross-validation provides a more reliable estimate of model performance than a single train-validation split, as it evaluates the model across multiple subsets of the data.

A grid search procedure was applied to identify the optimal combination of Random Forest hyperparameters. The search considered different values of the number of trees (`n_estimators`), maximum tree depth (`max_depth`), and minimum samples per leaf node (`min_samples_leaf`). Grid search with 5-fold cross-validation was applied only to the training set to tune the Random Forest hyperparameters. The temporal test set was kept separate and used only for final out-of-sample evaluation. This ensured that model selection was performed without exposing the model to future observations

The optimal parameter combination identified through cross-validation was:

- Number of trees (`n_estimators`) = 100
- Maximum tree depth (`max_depth`) = 20
- Minimum samples per leaf (`min_samples_leaf`) = 5

The final Random Forest model was subsequently trained using these optimal parameters and evaluated on the holdout test dataset. This process is repeated k times (in this study, $k = 5$) so that each fold serves as a validation set once (Kohavi, 1995).

This evaluation approach helps ensure that the model is tested fairly and that its results reflect its ability to make reliable predictions on new data rather than simply memorising the training

Hyperparameter	Values Tested
n_estimators	100, 200, 300
max_depth	10, 20, None
min_samples_leaf	1, 2, 5

Table 3.4: Hyperparameter Search Space

3.7 Evaluation Metrics

Several evaluation metrics were used to assess the performance of the model instead of relying only on accuracy. Since this study predicts multiple credit risk categories, a single metric cannot fully measure model performance. Using multiple metrics provides a better understanding of how well the model identifies different credit risk categories.

Therefore, the evaluation focuses not only on overall accuracy but also on the model's ability to correctly identify different levels of borrower risk.

3.7.1 Precision

Precision measures how often the model is correct when it predicts a particular credit risk category. In simple terms, it shows how reliable the model's predictions are for a given class

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP represents true positives and FP represents false positives.

3.7.2 Recall (Sensitivity)

Recall measures how many actual observations in a credit risk category are correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN represents false negatives.

TP represents True Positives

In this study, special attention is given to recall for high-risk borrowers because failing to identify them can lead to significant financial losses.

In imbalanced classification problems, accuracy alone can be misleading. A model may achieve high accuracy simply by predicting the largest category most of the time. For this reason, He and Garcia (2009) recommend using metrics such as recall and F1-score to provide a better assessment of model performance in imbalanced datasets. This supports the focus on recall in this study.

3.7.3 F1-Score

The F1-score combines **precision** and **recall** into a single measure.

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$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

It is useful because it considers both false positives and false negatives when evaluating model performance.

To account for class imbalance, this study uses the **macro-averaged F1-score**, which gives equal importance to all credit risk categories.

3.7.4 Confusion Matrix

A confusion matrix provides a detailed summary of the model's predictions across all credit risk categories.

It shows:

- **True Positives (TP):** Borrowers correctly assigned to their actual credit risk category.
- **False Positives (FP):** Borrowers incorrectly assigned to a credit risk category.
- **False Negatives (FN):** Borrowers whose true credit risk category was missed by the model
- **True Negatives (TN):** Borrowers correctly excluded from a credit risk category

The confusion matrix helps identify two important types of errors:

- **Type I Errors (False Positives)**
- **Type II Errors (False Negatives)**

In credit risk modelling, these errors have different consequences.

- Classifying a high-risk borrower as low risk can increase the risk of financial losses.
- Classifying a low-risk borrower as high risk may lead to unnecessary loan rejection.

For this reason, the confusion matrix is useful for understanding how reliable the model is in real-world credit risk decisions and portfolio management.

Evaluation Framework Summary

The combination of precision, recall, F1-score, and confusion matrix analysis provide a total evaluation of model performance.

The evaluation focuses on:

- Good performance across all credit risk categories
- Correct identification of high-risk borrowers
- Reliable performance despite class imbalance

This helps ensure that the model is not only accurate but also useful for real-world credit risk decision-making.

3.8 Model Explainability

To make the model easier to understand, this study uses **feature importance** and **SHAP**. Feature importance shows which variables have the greatest influence on model predictions, while SHAP explains how each variable contributes to those predictions (Lundberg & Lee, 2017).

Explainability is important in credit risk modelling because financial institutions need to understand why a model makes certain predictions. Since Random Forest can identify complex patterns in data, SHAP helps explain how the model reaches its decisions.

Two types of explanations are used in this study.

1 Global Feature Importance

SHAP values are first used to identify the most important variables affecting predictions across the entire dataset. This helps rank variables based on their overall influence on borrower credit risk outcomes.

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Examples include:

- Financial capacity (e.g., income and debt-to-income ratio)
- Loan characteristics (e.g., interest rate and loan amount)
- Behavioural indicators (e.g., delinquency history)

Understanding these factors helps explain why borrowers are classified into different credit risk categories and ensures that model predictions remain consistent with credit risk theory.

2. Local Prediction Interpretation

SHAP values are also used to explain individual predictions. For each borrower, SHAP shows how specific features increase or decrease the predicted credit risk category.

This helps explain why a borrower was classified into a particular risk group and identifies the factors that contributed most to the prediction. These borrower-level explanations are valuable because they improve understanding of individual credit risk outcomes and support better decision-making.

3.8.1 Role of Explainability in the Study

SHAP was used in this study for two main reasons:

- **Model transparency:** to explain how the Random Forest model makes its predictions.
- **Model validation:** to check whether the important features identified by the model are consistent with credit risk theory and financial reasoning.

The results of the SHAP analysis are presented in **Chapter 4**, where feature importance rankings and individual prediction explanations are used to interpret model behaviour.

3.9 Implementation Environment

All data analysis and modelling were carried out in Python using the following libraries:

- **Pandas** and **NumPy** for data cleaning and preparation.
- **Scikit-learn** for building, training, and evaluating the Random Forest model.
- **SHAP** for model explainability.
- **Matplotlib** and **Seaborn** for data visualisation.

The analysis was performed in a **Jupyter Notebook** environment, which allowed the modelling process to be developed, tested, and documented in a clear and organised way.

3.10 Summary

This chapter explained the methods used to predict borrower credit risk outcomes using the Lending Club dataset. The study followed a quantitative and data-driven approach, treating the problem as a supervised multi-class classification task.

The dataset was cleaned and transformed into features for modelling, while credit risk categories were created from loan outcomes.

An empirical transition matrix was developed as a benchmark, while Random Forest was used as the main predictive model. This model was trained using a temporal train-test split, **cross-validation**, **hyperparameter tuning**, and **class weighing** to address class imbalance. Model performance was evaluated using standard classification evaluation metrics.

SHAP explainability techniques were used to identify the most important factors influencing model predictions and to improve model transparency.

CHAPTER 4: Results and Discussion

4.1 Introduction

This chapter presents the results of the model developed in this study. Using the methodology described, It evaluates its ability to predict borrower credit risk outcomes.

The analysis started with an overview of the dataset and the preprocessing steps used to prepare the data. A stratified temporal sampling approach was applied to ensure that loans from different origination years were adequately represented, allowing the study to capture changes in credit conditions across different periods.

```
print("Initial dataset shape:", df.shape)
```

```
Initial dataset shape: (2260701, 151)
```

Figure 4.1 Dimensionality of the pre-processed pre-sampled dataset

```
print("Final modelling dataset shape:", df.shape)
```

```
Final modelling dataset shape: (77346, 156)
```

Figure 4.2 Dimensionality of the final modelling set

The final modelling dataset contains **77,346 loan records** covering the period from **2007 to 2018**. The larger number of loans in later years reflects increased lending activity and helps capture changes in credit conditions over time.

The main focus of the analysis is the **Random Forest model**. Model performance is evaluated using multi-class classification metrics, with particular attention given to identifying higher-risk borrowers.

To better understand model performance, a confusion matrix is used to see prediction accuracy across board. SHAP explainability techniques are also applied to identify the borrower and loan characteristics that have the greatest influence on model predictions.

The chapter concludes by discussing the results and explaining how they relate to the research objectives and existing credit risk literature.

4.2 Descriptive Statistics and Data Overview

This section provides an overview of the Lending Club dataset and highlights key borrower, loan, and credit risk characteristics. The aim is to understand the main patterns in the data before evaluating the prediction model.

Descriptive statistics such as the mean, median, standard deviation, and value ranges were calculated for key variables. Charts were also used to show the distribution of borrowers across different credit risk categories.

The results show that borrower characteristics vary across different risk levels. The analysis also shows that there are many more low-risk borrowers than high-risk borrowers. Also, the distribution of credit risk categories shows the presence of **class imbalance**, with low-risk borrowers greatly outnumbering high-risk borrowers.

Overall, the descriptive analysis provides useful background information for understanding the modelling results presented later in the chapter.

4.2.1 Distribution of Credit Risk States

The distribution of borrowers across different credit risk categories helps explain the overall risk profile of the loan portfolio.

This obvious difference between low-risk and high-risk borrowers is known as **class imbalance**. Although it reflects real-world lending behaviour, it can make prediction more difficult because models may focus too much on the larger categories.

In addition, **macro-averaged evaluation metrics** were used to ensure that model performance was measured fairly across all credit risk categories.

Distribution of Loans by Credit Grade

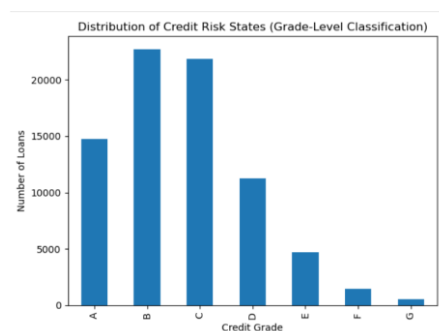


Figure 4.3: Distribution of Loans by Credit Grade

Source: Lending Club Dataset (Author's computation)

The distribution of loans across credit grades (A–G) shows that most loans fall within the middle grades (B–D). This suggests that the portfolio has a moderate overall level of credit

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risk. Less loans are found in grade **A**, while grades **E–G** contain a smaller but riskier group of borrowers

It is important to note that **loan grade is not the target variable being predicted**. Instead, loan grade represents the borrower's credit quality when the loan was issued and is used as an input feature in the model.

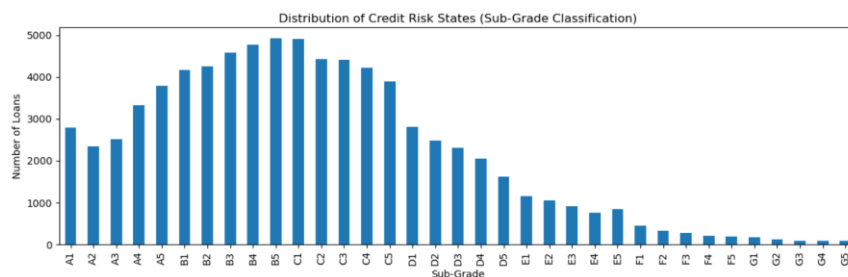


Figure 4.4: Distribution of Credit Risk States (Sub-Grade Classification)

Figure 4.4 provides a more detailed view of the portfolio by showing the distribution of loans across sub-grades. The results show that the number of loans generally decreases as credit quality becomes weaker, especially from **B4 to G5**. This suggests that fewer borrowers fall into the highest-risk categories.

4.2.2 Summary Statistics of Key Variables

Descriptive statistics were calculated for important borrower and loan variables used in the model.

The results show that:

- **Annual income** varies across borrowers, showing differences in financial capacity.
- **Debt-to-income ratio (DTI)** differs across borrowers, showing different levels of debt burden.

- **Loan amount** and **interest rate** also vary, shows differences in loan size and pricing.
- **Delinquency-related variables** are unevenly distributed, most borrowers have few or no missed payments, while a smaller group shows higher-risk repayment behaviour.

These patterns are common in consumer credit datasets and shine light on the importance of data preprocessing techniques such as handling outliers and preparing variables for modelling.

Variable	Count	Mean	Std. Dev	Min	Max
Loan Amount	77,346	14,503.19	9,077.32	500	40,000
Interest Rate	77,346	13.06	4.72	5.31	30.99
Annual Income	77,342	76,697.62	52,197.74	0	1,000,000
Debt-to-Income Ratio	77,285	17.87	8.82	0	59.98
Fico_avg	77,346	701.57	33.64	612	847.50
Revolving Balance	77,346	16,532.77	23,105.35	0	1,137,891
Delinquency History	77,317	0.29	0.85	0	42

Table 4.1: Summary Statistics of Key Credit Risk Variables

Source: Lending Club Dataset (Author's computation)

Table 4.1 shows the main borrower and loan characteristics in the dataset. The average loan amount is about **\$14,503**, moderate dispersion was observed. Average interest rate is **13.06%**, this suggests that many borrowers in the portfolio fall within a medium-risk range.

Borrower income varies widely, showing that the dataset contains borrowers with different financial situations. The average debt-to-income ratio is about **18%**, while the average FICO score of **702** suggests that most borrowers have moderate credit quality.

These results show that the dataset contains a wide range of borrower profiles, making it suitable for credit risk modelling.

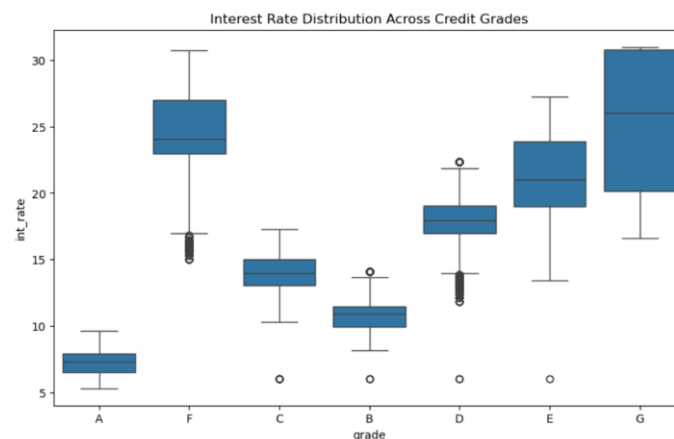


Figure 4.5: Distribution of Interest Rates Across Credit Grades

Figure 4.5 shows how interest rates change across different credit grades. The results show that borrowers with higher risk levels are usually charged higher interest rates. This is because lenders charge more interest to borrowers who are more likely to miss repayments or default. i.e considered riskier.

The results show that interest rates become more spread out in the higher-risk credit grades. This means that higher-risk borrowers are more different from one another in terms of their financial situations and repayment behaviour, making it harder to predict how they will repay their loans.

The results show that credit grades are closely linked to borrower risk. This is why credit grade information is useful in predicting credit risk outcomes.

4.2.3 Implications for Modelling

The descriptive analysis shows two important features of the dataset that affect model performance:

1. **Class Imbalance:** The dataset contains many more low-risk borrowers than high-risk borrowers. This can make it harder for the model to learn patterns from the smaller high-risk groups. To address this problem, **class weighting** was used during model training. This gives more importance to high-risk borrowers and helps the model identify them more effectively.
2. **Heterogeneity in Borrower Profiles:** Borrowers in the dataset have different financial situations and repayment behaviours. This suggests that the relationship between borrower characteristics and credit risk is complex. For this reason, a **Random Forest** model was used because it can identify complex patterns and relationships in the data.

Overall, the dataset reflects many of the characteristics found in real-world loan portfolios and provides a strong foundation for predicting borrower credit risk outcomes.

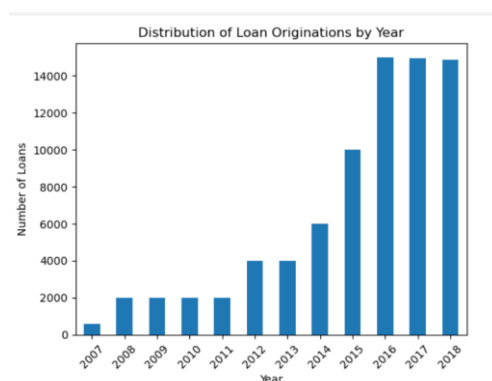


Figure 4.6 Distribution of Loan Originations by year

Figure 4.6 shows the number of loans issued each year. The results show that lending activity increased over time, with more loans being issued in later years. This growth may reflect changes in lending conditions and borrower demand over the study period. The differences across years also support the use of **stratified temporal sampling** in the modelling process.

4.3 Empirical Transition Matrix Analysis

4.3.1 Overview of Credit Risk Migration Dynamics

This section presents the empirical transition matrix created from the Lending Club dataset to better understand borrower credit risk outcomes.

In simple terms, the analysis estimates:

$$P(\text{Outcome State} \mid \text{Initial Credit Risk State})$$

Where:

- Initial state = **sub_grade (A1–G5)**
- Outcome = **Paid / Default / Active**

The results are presented in Table 4.2.

Sub-Grade	Paid	Default	Active
A1	0.491	0.019	0.491
A2	0.543	0.030	0.427
A3	0.535	0.035	0.429
A4	0.572	0.042	0.386
A5	0.579	0.058	0.363
B1	0.521	0.064	0.415
B2	0.54	0.075	0.385
B3	0.574	0.084	0.342
B4	0.542	0.096	0.362
B5	0.525	0.108	0.367

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C1	0.499	0.121	0.379
C2	0.505	0.131	0.364
C3	0.471	0.145	0.384
C4	0.462	0.144	0.394
C5	0.455	0.149	0.396
D1	0.475	0.185	0.34
D2	0.472	0.19	0.338
D3	0.468	0.19	0.342
D4	0.449	0.202	0.349
D5	0.463	0.222	0.316
E1	0.522	0.253	0.225
E2	0.498	0.258	0.244
E3	0.491	0.261	0.247
E4	0.487	0.267	0.246
E5	0.425	0.26	0.316
F1	0.476	0.3	0.224
F2	0.51	0.309	0.181
F3	0.463	0.338	0.199
F4	0.471	0.327	0.202
F5	0.454	0.351	0.196
G1	0.436	0.33	0.235
G2	0.404	0.43	0.167
G3	0.448	0.385	0.167
G4	0.407	0.442	0.151
G5	0.506	0.321	0.173

Table 4.2: Empirical Transition Probability Matrix (Sub-Grade Level)

Source: Author's computation

Table 4.2 shows the relationship between a borrower's initial credit quality and the final loan outcome. A clear pattern can be seen across the credit grades. As borrower risk increases, the probability of default also increases.

Borrowers in higher-risk sub-grades (**F and G**) are much more likely to default than borrowers in lower-risk sub-grades (**A to C**). This suggests that the credit grading system is effective at separating lower-risk borrowers from higher-risk borrowers. Although there are small differences between some neighbouring sub-grades, the overall pattern remains clear: borrowers with weaker credit grades generally have higher default rates. The relatively small

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number of loans in the **Active** category suggests that most loans in the dataset had already reached an outcome, making the estimated probabilities more reliable.

Reduced-Form Interpretation

It is important to note that this transition matrix does not show actual borrower movements over time. Instead, it provides a **reduced-form approximation of credit risk migration**, showing how a borrower's starting credit quality is related to their final loan outcome. This provides a practical way of analysing credit risk changes using the available data.

The results show that borrowers with higher risk today are more likely to have worse loan outcomes in the future, even though the dataset does not track their risk changes over time.

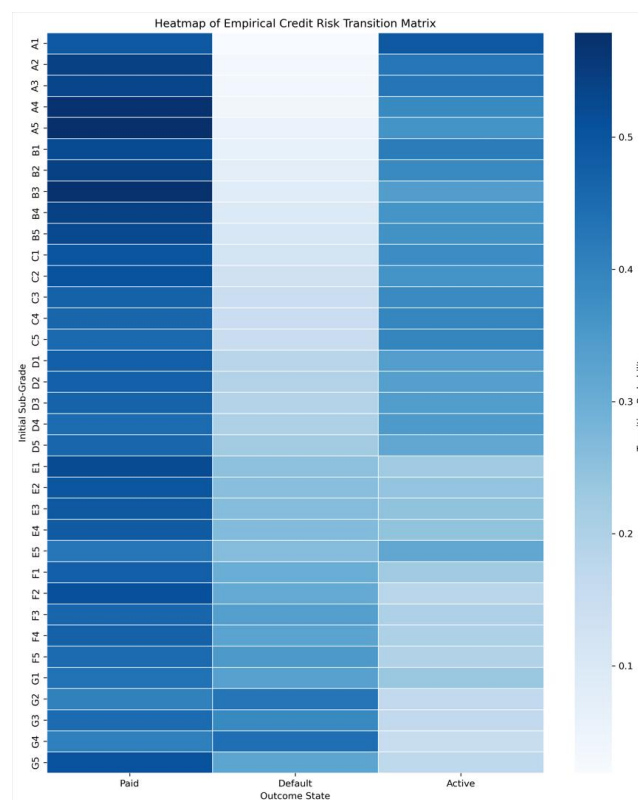


Figure 4.7: Heatmap of the Empirical Credit Risk Transition Matrix

To make the transition matrix easier to understand, it was displayed as a heatmap.
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This allows for clearer identification of patterns across sub-grade classifications.

The heatmap shows a clear increase in default risk as credit quality becomes weaker. Higher-risk sub-grades have darker values in the default column, indicating a greater chance of default.

At the same time, lower-risk grades show higher probabilities in the **Paid** category, reflecting stronger repayment behaviour.

Implementation Note

The transition matrix was computed using Python, with full implementation details provided in **Appendix C**.

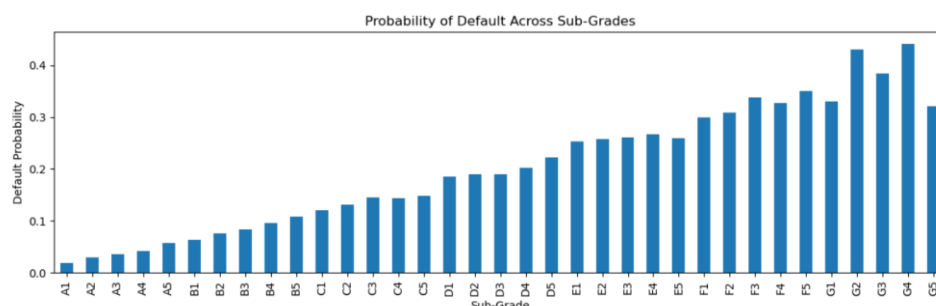


Figure 4.8: Probability of Default Across Sub-Grades

Source: Author's computation

To better understand the science of credit risk, the probability of default was examined across the different sub-grades. The results show that default risk generally increases as credit quality becomes weaker. This pattern shows that the credit grading system reflects borrower risk effectively and supports its use in predicting credit risk outcomes.

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4.4 Machine Learning Results

This section presents the results of the Random Forest model used to predict borrower credit risk outcomes. The model was trained using borrower financial information, repayment behaviour, and loan characteristics.

To preserve chronological ordering and to make the evaluation more realistic, a **temporal train-test split** was used. Loans issued up to **2015** were used for training, while loans issued after **2015** were used for testing. A temporal split was used because it better reflects how credit risk prediction works in the real world, where past borrower data is used to predict future outcomes.

4.4.1 Temporal Train-Test Split

Dataset	Total Observations	Low Risk Outcome	Intermediate Outcome	Adverse Outcome
Training Set (2007–2015)	32,596	25,811	1,257	5,528
Test Set (2016– 2018)	44,750	13,528	27,099	4,123

Table 4.3: Temporal Train-Test Split

Source: Author's computation

Table 4.3 shows how the dataset was divided into training and testing sets. The training set contains loans issued between **2007 and 2015**, while the test set contains loans issued between **2016 and 2018**. This approach allows the model to learn from earlier loans and be tested on newer loans, providing a more realistic measure of performance.

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4.4.2 Cross-Validation and Hyperparameter Optimisation Results

Initial tests using a single train-test split produced a modest performance. To improve the model, **5-fold cross-validation** was used during training. This allowed the model to be tested on different parts of the data, giving a more reliable measure of performance. The final Random Forest model was then selected using cross-validation and hyperparameter optimization

Hyperparameter optimisation was performed using Grid Search combined with 5-fold cross-validation. This process systematically evaluated multiple Random Forest configurations to identify the model that achieved the best predictive performance on the training dataset.

A total of 18 hyperparameter combinations were assessed, resulting in 90 model fits across the five validation folds. The results are presented in Table 4.4.

The highest cross-validation score was achieved using a maximum tree depth of 20, a minimum leaf size of 5, and 100 decision trees. This configuration was selected as the final Random Forest model and subsequently evaluated on the holdout test dataset.

The use of cross-validation improves model robustness by reducing the likelihood that model performance is influenced by a single train-validation split. It also provides greater confidence that the model can generalise to unseen borrower observations.
data.

Rank_test_scores1	Rank	Param_n_estimators	Param_max_depth	Param_mean_samples_leafs	mean_test_score
1	10	100	20	5	0.989560
2	16	100	None	5	0.989110
3	11	200	20	5	0.989075
4	17	200	None	5	0.988963
5	3	200	10	2	0.988948
6	5	200	10	5	0.988935
7	4	100	10	5	0.988329
8	14	100	None	2	0.987879
9	0	100	10	1	0.987845
10	2	100	10	2	0.987838

Table 4.4: Top Hyperparameter Combinations from 5-Fold Cross-Validation

Table 4.4 presents the highest-performing Random Forest hyperparameter combinations identified through 5-fold cross-validation. The optimal configuration consisted of a maximum tree depth of 20, a minimum leaf size of 5, and 100 trees, achieving the highest mean cross-validation score of 0.9896. The small differences between the top-ranked models indicate that the Random Forest model exhibited stable performance across multiple parameter configurations.

Hyperparameter	Optimal Value
Number of Trees (n_estimators)	100
Maximum depth (max_depth)	20
Minimum sample per leaf (min_samples_leaf)	5
Class Weight	Balanced
Cross Validation	5 - fold

Table 4.5: Optimal Hyperparameters Selected by Grid Search

From Table, 4.5, we see that the best-performing Random Forest model used 100 trees, a maximum depth of 20, and a minimum leaf size of 5. This configuration achieved the highest mean cross-validation score of 0.9896 and was therefore selected for the final model evaluation

Metric	Value
Mean Cross-Validation Score	0.9896
Standard Deviation	0.0030
Number of folds	5
Hyperparameter Combinations Tested	18
Total Model Fits	90

Table 4.6: Cross-Validation Performance of the Optimised Random Forest Model

Source: Author's Computation

Table 4.6 presents the results of the 5-fold cross-validation and hyperparameter optimisation process. The optimal Random Forest model achieved a mean cross-validation score of 0.9896

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with a standard deviation of 0.0030, indicating stable performance across the validation folds. The best-performing configuration consisted of 100 trees, a maximum tree depth of 20, and a minimum leaf size of 5. These results suggest that the model generalises well and is not overly sensitive to variations in the training data.

4.4.3 Classification Performance

Following hyperparameter optimisation and cross-validation, the final model was evaluated on the holdout test dataset to assess its predictive performance under unseen data conditions. The predictive performance of the Random Forest model is evaluated using a range of classification metrics.

	Precision	Recall	F1-Score	Support
Adverse Outcome	0.58	0.96	0.72	4123
Intermediate Outcome	1.00	0.66	0.79	27099
Low Risk Outcome	0.67	1.00	0.80	13528
Accuracy			0.79	44750
Macro avg	0.75	0.87	0.77	44750
Weighted avg	0.86	0.79	0.79	44750

[[3950 1 172]

[2916 17752 6431]

[0 0 13528]]

Table 4.7: Classification Report for Credit Risk Outcome Prediction

Source: Author's computation

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Table 4.7 shows how well the improved Random Forest model predicted borrower risk on the holdout test data. The model achieved an overall accuracy of **79%**, meaning it correctly classified most borrowers after hyperparameter optimisation and cross-validation.

The model performed very well in identifying **Adverse Outcomes**, with a recall score of **0.96**. This means that the model correctly identified most high-risk borrowers.

The model also achieved strong performance for **Low-Risk Outcomes**, with a recall score of **1.00**, meaning that almost all low-risk borrowers were correctly classified. For **Intermediate Outcomes**, the model achieved high precision of **1.00**, although recall was lower at **0.66**, showing that some intermediate-risk borrowers were placed in the wrong risk category.

The macro-average F1-score of **0.77** shows that the model performed reasonably well across all three credit risk categories.

The improvement in model performance was caused by the use of cross-validation during model development. Rather than relying on a single train-test split, the model was evaluated across multiple folds of the training data. This helped identify better hyperparameter settings, reduced the risk of overfitting, and improved the model's ability to make accurate predictions on unseen data.

Overall, the results show that the optimised Random Forest model was able to learn useful patterns from borrower data and predict credit risk outcomes effectively.

4.4.4 Confusion Matrix Analysis

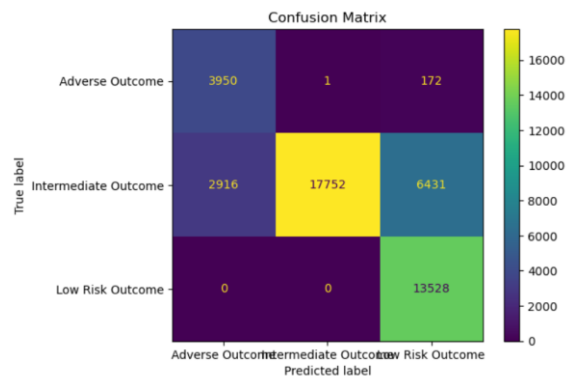


Figure 4.9: Confusion Matrix for Credit Risk Outcome Prediction

Source: Author's computation

Figure 4.9 shows the confusion matrix for the optimised Random Forest model. The confusion matrix provides a clearer view of how the model classified borrowers across the credit risk categories.

The model correctly identified **3,950 out of 4,123 high-risk borrowers**, showing that it was very good at finding borrowers who were more likely to have repayment problems.

For **intermediate-risk borrowers**, the model correctly classified **17,752 out of 27,099 cases**.

Some borrowers in this group were placed in the wrong category, making this the hardest group to predict.

For **Low Risk Outcomes**, the model correctly classified all **13,528 cases**, showing very strong performance.

Overall, the confusion matrix shows that the optimised model was very good at identifying both low-risk and high-risk borrowers, while intermediate-risk borrowers were more difficult to classify accurately.

4.4.5 Feature Importance Analysis

After evaluating model performance, the next step is to identify the factors that have the greatest influence on the model's predictions.

Feature importance was extracted from the Random Forest model.

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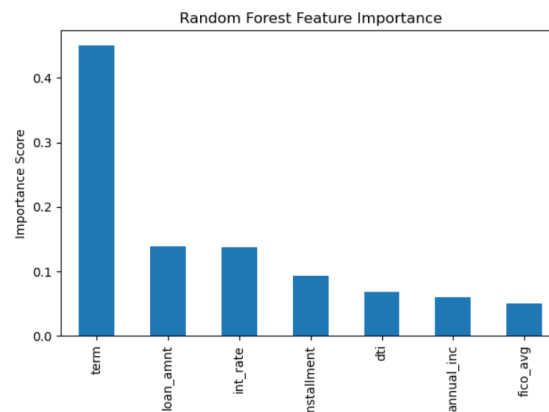


Figure 4.10: Random Forest Feature Importance

Source: Author's computation

Figure 4.10 shows the variables that contributed most to predicting borrower credit risk outcomes. The results show that borrower financial information, repayment behaviour, and loan characteristics are important predictors of credit risk.

The most important variables include:

- loan term
- interest rate

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- debt-to-income ratio
- FICO score
- revolving balance

These variables are closely linked to borrower financial strength, repayment ability, and overall credit quality.

Among all variables, **loan term** was the most important predictor. This suggests that the length of a loan provides valuable information about future credit risk outcomes.

Overall, the results confirm that the selected features are useful for predicting borrower credit risk.

4.4.6 SHAP-Based Model Explainability

To better understand how the model makes predictions, **SHAP** was used. SHAP helps explain how different features influence the model's predictions. Unlike traditional feature importance measures, SHAP values will show both the magnitude and class-specific influence of each feature.

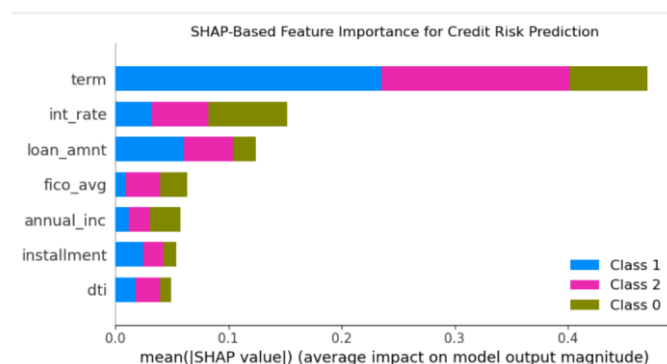


Figure 4.11: SHAP Summary Plot

Source: Author's computation

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Figure 4.11 shows the importance of different features in predicting credit risk outcomes.

To enhance model interpretability, SHAP analysis was applied to see how each and every feature add to model predictions. The results show that borrower financial and repayment-related variables play an important role in the model's predictions. Some of the most important features include:

- Interest rate
- FICO score
- Debt-to-income ratio (DTI)
- Loan term

The results suggest that borrowers with higher interest rates and lower credit scores are generally more likely to experience negative credit outcomes. On the other hand, borrowers with stronger financial profiles are more likely to fall into lower-risk categories.

The SHAP analysis supports the earlier feature importance results and provides a clearer understanding of how different variables influence model predictions. Most importantly, it improves transparency in the decision-making process.

4.5 Ethical Considerations and Model Limitations

Using machine learning for credit risk prediction raises important ethical and regulatory considerations, related to fairness, transparency, and responsible decision-making.

One concern is **algorithmic bias**. Credit data may contain patterns from past lending decisions that could affect how the model treats different groups of borrowers. Although this study does not use sensitive personal information directly, some variables may still indirectly reflect these differences.

Another challenge is **class imbalance**. To mitigate, **class weighting** was used during model training. However, some prediction errors still occur between neighbouring risk categories.

To improve transparency, **SHAP explainability techniques** were used to show transparency in decision making.

The study also uses a publicly available and anonymised dataset, which helps protect borrower privacy. However, any real-world application of the model would need to follow data protection and privacy regulations.

Finally, the study has some limitations. The dataset does not provide fully observed longitudinal borrower transitions, so therefore, the modelling framework represents a reduced-form approximation of credit risk migration.

Overall, the model shows useful predictive ability, but real-world use would require careful attention to fairness, transparency, regulatory requirements, and responsible model management.

4.6 Key Findings and Insights

This section summarizes the main findings from the descriptive analysis, transition matrix, and Random Forest model, it shone light on key drivers of borrower credit risk outcomes.

1. Credit Grades Are Linked to Loan Outcomes

The results show that borrowers with better credit grades are more likely to repay their loans, while borrowers with weaker credit grades are more likely to default.

Key Insight: Credit grades provide useful information about borrower risk.

2. Borrower Risk Changes Across Different Risk Levels

Higher-risk borrowers are more likely to experience negative loan outcomes, while lower-risk borrowers are generally more stable.

Key Insight: Credit risk can be better understood by looking at different risk levels rather than only focusing on default.

3. The Random Forest Model Identified Important Risk Patterns

The Random Forest model was able to identify important patterns in borrower credit risk and predict credit risk outcomes reasonably well.

Key Insight: Tree-based machine learning models works well for modelling nonlinear relationships and complex interactions in data.

4. Class Imbalance Affected Model Performance

The dataset contained many more low-risk borrowers than high-risk borrowers. This made it harder for the model to identify some high-risk cases.

Key Insight: Class imbalance remains an important challenge in credit risk modelling.

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5. Financial Variables Were Important Predictors

Variables such as interest rate, loan term, FICO score, and debt-to-income ratio had the greatest influence on model predictions.

Key Insight: Borrower financial strength plays a major role in credit risk outcomes.

6. SHAP Improved Understanding of the Model

SHAP helped explain how different variables influenced the model's predictions.

Key Insight: Explainability tools make models easier to understand and trust.

7. Balancing Accuracy and Risk Detection

Improving the model's ability to identify high-risk borrowers sometimes reduces overall accuracy.

Key Insight: In credit risk modelling, identifying risky borrowers is often more important than achieving the highest possible accuracy.

4.7 Summary

This chapter presented the results of the study using descriptive analysis, transition matrix analysis, and machine learning. The findings showed a clear relationship between borrower credit quality and loan outcomes, with stronger credit grades generally associated with lower risk. The Random Forest model achieved good predictive performance, while feature importance and SHAP analysis confirmed that its predictions were driven by meaningful financial variables. Overall, the results suggest that combining transition matrix analysis with machine learning provides a useful framework for understanding and predicting borrower credit risk outcomes.

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CHAPTER 5: Discussion and Interpretation

5.1 Introduction

This chapter discusses the main findings of the study and explains what they mean in relation to the research objectives and existing credit risk literature. It evaluates how well the proposed framework predicts borrower credit risk outcomes. The findings support the idea that borrower risk changes gradually over time and can be analysed using credit risk migration approaches. This aligns with established portfolio credit risk frameworks such as CreditMetrics (Gupton, Finger, & Bhatia, 1997).

The discussion focuses on three main areas.

First, it examines how the transition matrix helps explain differences in borrower risk beyond simply looking at default or non-default.

Second, it evaluates how well the Random Forest model predicts borrower credit risk outcomes using borrower and loan information.

Third, it discusses how SHAP explainability techniques help make machine learning predictions easier to understand and interpret.

These findings are consistent with prior research highlighting the challenges of predicting rare events in financial datasets (Lessmann et al., 2015).

The chapter also considers the practical value of the findings. In addition, the limitations of the study are discussed, including issues related to the model.

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Finally, recommendations are provided for future research and the practical use of data-driven credit risk models.

5.2 Discussion and Interpretation of Key Findings

The findings of this study provide useful insights into how machine learning and transition analysis can be used to predict borrower credit risk. The results show that borrower risk cannot be fully understood by looking only at whether a borrower defaults or not. Borrowers exist at different levels of risk, which provide valuable information for credit risk analysis.

The transition matrix showed a clear relationship between borrower credit grades and loan outcomes. Borrowers with stronger credit grades were more likely to repay their loans, while borrowers with weaker grades were more likely to experience negative outcomes. Although the dataset does not track borrower risk changes over time, these results show that a reduced-form approach can still provide useful insights into credit risk patterns.

The Random Forest model achieved an accuracy of 78.7% following hyperparameter optimisation using Grid Search and 5-fold cross-validation. The model successfully identified relationships between borrower financial characteristics, repayment behaviour, and credit risk outcomes. By systematically evaluating multiple hyperparameter combinations, the optimisation process identified a Random Forest configuration that improved predictive performance and reduced the risk of overfitting. This demonstrates the importance of model tuning in credit risk modelling, where reliable prediction is essential for effective risk assessment and decision-making.

These findings are consistent with previous studies showing that machine learning models can effectively identify complex patterns in borrower data, especially in large and detailed datasets. Bussmann et al. (2021) and Lessmann et al. (2015) show that machine learning models often outperform traditional approaches in capturing complex borrower behaviour patterns.

The dataset contained many more low-risk borrowers than high-risk borrowers, which made it harder for the model to identify high-risk cases. This was rectified by; **class weighting** which was used during training.

The confusion matrix also showed that the differences between neighbouring risk categories were often small. This suggests that borrower risk changes gradually rather than fitting into clearly separated groups.

The feature importance analysis showed the variables below, were the most important factors affecting credit risk outcomes.

- **Loan Term:** Longer loan terms may increase exposure to adverse economic conditions and repayment uncertainty, explaining their strong predictive influence
- **Interest rate:** Higher interest rates are often assigned to borrowers who are considered riskier. Therefore, it serves as an indicator of the likelihood of adverse outcomes.
- **Fico Score:** This reflects a borrower's creditworthiness based on past credit behaviour. Borrowers with higher FICO scores are more likely to repay their loans,
- **Debt-to-income ratio:** Higher ratios may indicate greater financial pressure and a reduced ability to repay debt, increasing the likelihood of adverse credit outcomes.

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This makes sense because these variables reflect a borrower's financial strength, repayment ability, and credit history

The SHAP analysis showed how each variable influenced individual borrower predictions. While feature importance identified the most important variables overall. This highlights the value of explainability tools, especially in financial applications where model decisions need to be clear and easy to understand.

Overall, the findings suggest that combining transition analysis with machine learning provides a useful way to analyse borrower credit risk, it shows how data-driven methods can provide a more detailed view of credit risk than traditional default prediction models.

5.3 Practical Implications

5.3.1 Implications for Financial Institutions

The findings show the importance of looking beyond simple default prediction when assessing borrower risk. The transition analysis showed a clear relationship between borrower credit grades and loan outcomes. This means that financial institutions can identify signs of increasing risk before a borrower actually defaults.

The findings suggest that financial institutions can strengthen credit risk monitoring by incorporating predictive analytics into existing risk management processes. Machine learning models can support earlier identification of borrowers showing signs of deterioration, allowing institutions to take proactive measures before losses occur.

In practice, these insights can improve portfolio monitoring, borrower segmentation, risk-based pricing, and credit decision-making. The results also highlight the importance of reliable and transparent risk assessment. Predictive models can be used as complementary tools to support more informed and forward-looking credit risk management.

5.3.2 Implications for Risk Management Practice

The findings show that credit risk assessment should focus not only on overall model accuracy but also on how well the model identifies high-risk borrowers. In practice, missing a high-risk borrower can lead to greater financial losses than making small prediction errors in lower-risk categories.

The study also shows that borrower and loan characteristics can provide useful information for assessing credit risk. Understanding the factors linked to higher risk can help financial institutions make better lending and portfolio management decisions.

These insights can support:

- better borrower screening
- more effective portfolio monitoring
- earlier identification of potential problem loans
- more informed credit risk decisions

The use of explainability tools also helps users understand how the model reaches its predictions, making the decision-making process more transparent and easier to justify.

Overall, the findings suggest that effective credit risk management requires a balance between model performance, transparency, and practical decision-making needs.

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5.3.3 Implications for Regulators and Policy Frameworks

The modelling framework used in this study supports a more forward-looking approach to credit risk assessment by focusing on signs of borrower risk before default occurs. This is consistent with modern regulatory frameworks that encourage early identification of credit deterioration.

The study also highlights the importance of model transparency. As machine learning becomes more widely used in financial services, regulators increasingly expect models to be understandable, explainable, and properly monitored.

The results further show that model performance can be affected by data quality and class imbalance. This highlights the need for strong model validation and regular performance monitoring to ensure that risk models remain reliable.

More broadly, the findings suggest that successful use of machine learning in credit risk management depends on:

- high-quality data
- effective model governance
- skilled analytical teams
- strong regulatory compliance

When supported by these factors, machine learning can help organisations build more proactive and data-driven credit risk management systems.

5.4 Limitations of the Study

Like all research studies, this study has some limitations.

First, the dataset does not track borrowers over multiple time periods. As a result, the study estimates credit risk outcomes rather than observing actual changes in borrower risk over time.

Second, the analysis is based on information available in the Lending Club dataset. Important factors such as economic conditions, unemployment levels, and interest rate changes were not included and may also influence borrower credit risk.

Third, the study focuses on one machine learning model (Random Forest). Although this model performed reasonably well, other machine learning techniques may produce different results.

Finally, the findings are based on data from one lending platform. Because lending practices and borrower behaviour can differ across institutions and countries, the results may not apply equally to all credit markets.

These limitations should be considered when interpreting the findings and provide opportunities for future research.

5.5 Recommendations and Future Research

Based on the findings of this study, several recommendations can be made for future research.

First, future research should use longitudinal credit datasets that track borrower behaviour over time. This would provide a more complete picture of how borrower risk changes and improve the modelling of credit risk dynamics.

Second, further work should explore advanced imbalance-handling techniques. In this study, we explored class weighting to improve model performance, alternative approaches and cost sensitive approaches like synthetic sampling methods (e.g., SMOTE) could further enhance the detection of adverse outcomes (Chawla et al., 2002). Improving identification of high-risk borrowers remains very important in credit risk modelling.

Third, future research should examine the impact of economic factors such as interest rates, inflation, unemployment, and economic growth on borrower credit risk.

Fourth, future studies should test different machine learning models to see which one performs best. While this study used Random Forest, other models such as Gradient Boosting and Neural Networks could also be explored to see whether they provide better predictions.

Fifth, further research should extend the application of XAI techniques. While SHAP was used in this study, additional methods such as LIME (Local Interpretable Model-Agnostic Explanations), Partial Dependence Plots (PDPs), and Individual Conditional Expectation (ICE) plots may provide further insights into model behaviour and borrower predictions.

Finally, future work should examine the practical integration of machine learning models into institutional credit risk frameworks, including how predictive models can complement traditional tools such as transition matrices, regulatory capital frameworks, etc

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Overall, these recommendations show light on opportunities to improve credit risk modelling through quality data, additional variables, and more advanced analytical approaches.

5.6 Implementation Considerations and Constraints

Although the results of this study are encouraging, there are several practical challenges that may be faced when using similar models in real-world credit risk management:

5.6.1 Data Availability and Quality Constraints

A predictive model is only as good as the data used to build it. The effectiveness of predictive credit risk models depends on the availability of high-quality, structured, and consistent data. Financial institutions need accurate, complete, and up-to-date borrower information to produce reliable results. Poor data quality, missing records, and inconsistent reporting can reduce model performance and lead to less reliable predictions.

In the case of this study, the absence of fully observed longitudinal data challenged the model's true credit risk transitions over time.

5.6.2 Model Deployment and Operational Integration

Many financial institutions already use established credit risk systems and processes.

Introducing machine learning models may require changes to existing systems, workflows, and reporting procedures.

Practical deployment would require:

- existing credit scoring systems
- risk reporting frameworks

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- decision-making workflows

As a result, successful implementation often depends on having the right technology, infrastructure, and support systems in place.

5.6.3 Organizational Capability

Successfully using machine learning models requires the right people, skills, and processes.

Financial institutions need staff with knowledge of data science, risk management, and model governance to develop, monitor, and maintain these models.

5.6.4 Model Interpretability and Governance

Any model used in credit risk management must meet regulatory and business requirements.

This means that model results should be understandable, properly documented, and regularly reviewed to ensure they remain reliable and fair. Organisations must therefore balance predictive performance with transparency and accountability

Institutions must therefore establish governance structures that support:

- model validation
- ongoing monitoring
- accountability in model-driven decisions

5.6.5 Cost and Resource Considerations

Implementing machine learning models can require major investment in data management, technology infrastructure, and ongoing model maintenance. For some organisations, these costs may limit large-scale adoption, especially where resources are constrained.

5.6.6 Regulatory and Compliance Considerations

The use of machine learning in credit risk management must follow regulatory requirements and industry standards. Financial institutions need to ensure that their models **are transparent, fair, well documented, easy to review and explain.**

As machine learning becomes more common in finance, regulators are paying closer attention to how these models are developed and used. A major challenge is finding the right balance between strong model performance and regulatory compliance.

5.6.7 Model Risk

A challenge is **model risk**. Models are built using past data, but future economic conditions may be different from the past. As a result, model performance can change over time.

To keep models reliable, institutions should:

- monitor model performance regularly
- update models when necessary
- test how models perform under different economic conditions

These would ensure that models remain accurate and useful in real-world decision-making.

Overall, the successful use of machine learning in credit risk management depends not only on model performance but also on data quality, system integration, skilled personnel, effective governance and regulatory constraints. These factors are essential for ensuring that predictive models provide reliable support for credit risk decision-making.

5.7 Summary

This chapter discussed the main findings of the study and what they mean for credit risk management.

The results showed that borrower risk differs across credit grades, showing the importance of looking past simple default prediction when assessing credit risk. The Random Forest model was able to identify important patterns in borrower risk and predict credit risk outcomes with reasonable success.

The chapter also discussed how the findings can support better credit risk management and decision-making in financial institutions.

In addition, the study's limitations and practical challenges were highlighted, including data availability, model implementation, and organisational readiness.

Finally, recommendations were provided for future research, including the use of richer datasets, additional modelling techniques, and stronger integration of data-driven methods into credit risk management.

Overall, this chapter brought together the key findings, implications, limitations, and recommendations of the study, providing a bridge to the final conclusions presented in Chapter 6.

CHAPTER 6: Conclusion and Recommendations

6.1 Overview of the Study

This study explored how data-driven methods can be used to analyse borrower credit risk within loan portfolios. The study was motivated by the limitations of traditional models, which only focus on whether a borrower defaults or not.

Using the Lending Club dataset, the study combined transition matrix analysis with a Random Forest classification model to analyse credit risk outcomes. The aim was to move beyond simple default prediction and provide a more detailed understanding of borrower risk.

The study therefore contributes to credit risk analytics by demonstrating how transition analysis, machine learning, and explainability tools can be used together to support credit risk analysis and decision-making.

6.2 Summary of Key Findings

The study found a clear relationship between borrower credit grades and loan outcomes. Borrowers with stronger credit grades were more likely to repay their loans, while those with weaker grades were more likely to experience negative outcomes. This shows that credit grades help explain borrower risk better than looking only at default.

The Random Forest model achieved strong predictive performance after hyperparameter optimization and cross-validation. The confusion matrix showed that some borrower risk groups share similar characteristics, making them overlap and more difficult to distinguish.

Finally, feature importance and SHAP analysis identified some of the most important factors influencing borrower risk. These findings are consistent with credit risk theory and make the model easier to understand.

6.3 Conclusions Drawn from the Study

The study finalizes that credit risk cannot be fully captured through binary default modelling alone. It suggests that borrower risk is better understood as a range of risk levels rather than simply default or non-default.

The study demonstrates that machine learning can provide useful insights into borrower credit risk and support more detailed risk assessment. The Random Forest model, in particular, was able to capture nonlinear relationships between borrower characteristics and realised risk categories. The use of Grid Search and 5-fold cross-validation further improved the reliability of the modelling framework by identifying an optimal Random Forest configuration.

This study also demonstrate that predictive performance can be challenged by class imbalance and overlapping borrower profiles, leading to difficulty in distinguishing alike risk categories.

The study further concludes that explainability is essential in credit risk modelling, it highlights the importance of using transparent and explainable models so that predictions can be understood and trusted, thereby strengthening the interpretability of the model.

Lastly, the study demonstrates that meaningful credit risk insights can be derived from non-longitudinal loan data, providing a practical alternative where traditional migration modelling is not feasible

Overall, the study shows that combining traditional credit risk analysis with machine learning can provide a more complete understanding of borrower risk and support better credit risk management decisions. Worthy to note is that, it does not replace traditional transition-based analysis, but can enhance predictions.

6.4 Limitations of the Study

This study has some limitations.

The dataset does not track borrowers over time, making it difficult to observe actual changes in credit risk.

The study also uses data from a single lending platform and focuses on one machine learning model, which may limit how widely the findings can be applied.

6.5 Final Reflection

This study shows that data-driven methods can provide a more detailed understanding of borrower credit risk than traditional default prediction alone. By combining transition analysis, machine learning, and explainability techniques, the study developed a practical framework for analysing credit risk using real-world loan data.

The findings demonstrate the value of data-driven approaches in supporting more informed, transparent, and forward-looking approaches to credit risk and portfolio management, while also providing a foundation for future research and model development.

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Appendices

Appendix A: Dataset Variables Used in the Study

Variable	Description
loan_amnt	Total amount borrowed by the borrower
int_rate	Interest rate assigned to the loan
term	Loan repayment term
annual_inc	Borrower's reported annual income
dti	Debt-to-income ratio
fico_avg	Average FICO score derived from FICO range low and high
revol_bal	Borrower's revolving credit balance
revol_util	Revolving credit utilisation rate
delinq_2yrs	Number of delinquencies in the past two years
open_acc	Number of open credit accounts
total_acc	Total number of credit accounts
sub_grade	Lending Club credit sub-grade assigned at origination
loan_status	Observed loan performance outcome
migration_class	Derived credit risk outcome class used as the target variable

Appendix B: Data Preprocessing Pipeline

```
import pandas as pd
import numpy as np

# Load Lending Club dataset
df = pd.read_csv("accepted_2007_to_2018Q4.csv")

samples = []

for chunk in pd.read_csv(file_path, chunksize=100000, low_memory=False):

    chunk['issue_d'] = pd.to_datetime(
        chunk['issue_d'],
        format='%b-%Y',
        errors='coerce'
    )

    chunk = chunk[chunk['issue_d'].notna()].copy()
    chunk['issue_year'] = chunk['issue_d'].dt.year

    for year, grp in chunk.groupby('issue_year'):
        take = min(len(grp), 2000)
        samples.append(grp.sample(n=take, random_state=42))

df = pd.concat(samples, ignore_index=True)
```

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```

# Balance observations across years
df = (
    df.groupby('issue_year', group_keys=False)
    .apply(lambda x: x.sample(n=min(len(x), 15000), random_state=42))
    .reset_index(drop=True)
)

print(df.shape)
print(df['issue_year'].value_counts().sort_index())

# Create average FICO score
df['fico_avg'] = (
    df['fico_range_low'] + df['fico_range_high']
) / 2

# Handle missing values
numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns
df[numeric_cols] = df[numeric_cols].fillna(df[numeric_cols].median())

categorical_cols = df.select_dtypes(include=['object']).columns
df[categorical_cols] = df[categorical_cols].fillna("Missing")

```

Appendix C: Python Code for Empirical Transition Matrix Construction

```

import pandas as pd

# Create transition matrix using initial sub-grade and observed outcome
transition_counts = pd.crosstab(
    df['sub_grade'],
    df['outcome_state']
)

# Convert counts to probabilities
transition_matrix = transition_counts.div(
    transition_counts.sum(axis=1),
    axis=0
)

print(transition_matrix)

# Export transition matrix
transition_matrix.to_csv(
    'transition_matrix.csv'
)

Python Code for Transition Matrix Heatmap
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(12,8))

sns.heatmap(
    transition_matrix,
    annot=True,
    fmt='.2f',
    cmap='YlGnBu'
)

plt.title(
    'Heatmap of Empirical Credit Risk Transition Matrix'
)

```

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```
plt.xlabel('Outcome State')
plt.ylabel('Initial Sub-Grade')

plt.tight_layout()
plt.show()
```

Appendix D: Grid Search and Cross-Validation Results

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV

param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [10, 20, None],
    'min_samples_leaf': [1, 2, 5]
}

rf = RandomForestClassifier(
    class_weight='balanced',
    random_state=42,
    n_jobs=-1
)

grid_search = GridSearchCV(
    estimator=rf,
    param_grid=param_grid,
    cv=5,
    scoring='f1_macro',
    n_jobs=-1,
    verbose=1
)

grid_search.fit(X_train, y_train)

best_rf = grid_search.best_estimator_

print("Best Parameters:")
print(grid_search.best_params_)
```

Appendix E: Classification Report and Confusion Matrix

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

y_pred = best_rf.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

Appendix F: SHAP Explainability Code

```
import shap
```

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```
# Create SHAP explainer for Random Forest model
explainer = shap.TreeExplainer(best_rf)

# Calculate SHAP values
shap_values = explainer.shap_values(X_test)

# Generate SHAP summary plot
shap.summary_plot(
    shap_values,
    X_test
)
```

The complete Python implementation is available upon request.